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Contents

1	Equipment Module Class Design – EM_FR_CRY (Crystallization)	6
1.1	Functional Description	6
1.2	Failure Monitors	6
1.3	Instances	7
1.4	Control Logic	9
1.4.1	Command 00 – Hold	9
1.4.2	Command 01 – Filling	9
1.4.3	Command 02 – Homogenization	11
1.4.4	Command 03 – Reheat	12
1.4.5	Command 04 – Crystallization	14
1.4.6	Command 05 – Holding	16
1.4.7	Command 06 – Ready For Filtration	17
1.4.8	Command 08 – Filtration	19
1.4.9	Command 09 – Reject	20
1.4.10	Command 11 – STOP	24
1.4.11	Command 12 – INITIAL	24
1.5	Additional Logic/Sequence	25
1.5.1	Intelligent Cooling System - FR_ICS_CRY(n)	26
1.5.2	Auto-Jump Segment Sequence - AUTOJUMP_SEGMENT	26
1.5.3	Cooling Change-over Temperature - CCT_SEQUENCE	26
1.5.4	Holding Sequence - HOLDING_SEQUENCE	27
1.5.5	Agitator Speed for Filling - FILL_AGI_SPEED	27
1.5.6	Circulation Pump Start Sequence – CIRCULATION_PUMP	28
1.6	Parameters	28
1.7	Configurable Parameters	30
1.8	Named Set	30
2	Equipment Module Class Design – EM_FR_FLT (Filtration)	31
2.1	Functional Description	31
2.2	Failure Monitors	31
2.3	Instances	33
2.4	Control Logic	34
2.4.1	Command 00 – Hold	34
2.4.2	Command 01 – Filter Press Closing	34
2.4.3	Command 02 – Feed Pump Start	35
2.4.4	Command 03 – Adjust Sqz Pressure For Core Blow	38
2.4.5	Command 04 – Squeezing	39
2.4.6	Command 05 – Core Blow 1	41
2.4.7	Command 06 – Filtrate Blowing 3	43
2.4.8	Command 07 – Squeeze Oil	44
2.4.9	Command 08 – Adjust Sqz Pressure for Filtrate Blow 1	45
2.4.10	Command 09 – Cake Discharge	47
2.4.11	Command 10 - Core Blow 2	48
2.4.12	Command 11 – Adjust Sqz Pressure for Filtrate Blow 2	50

2.4.13	Command 12 – Filtrate Blowing 4.....	51
2.4.14	Command 13 – STOP	52
2.4.15	Command 14 – Initial.....	53
2.4.16	Command 15 – Filtrate Blowing 1.....	53
2.4.17	Command 16 - Filtrate Blowing 2	54
2.5	Additional Logic/Sequence	56
2.5.1	Fill TK241	56
2.5.2	Fill TK261	57
2.6	Parameters.....	57
2.7	Configurable Parameters.....	60
2.8	Named Set	60
3	Equipment Module Class Design – EM_FR_WSH (Washing)	61
3.1	Functional Description	61
3.2	Failure Monitors	61
3.3	Instances.....	62
3.4	Control Logic.....	63
3.4.1	Command 00 – Hold.....	63
3.4.2	Command 01 – Preheat.....	63
3.4.3	Command 02 – Washing 1	65
3.4.4	Command 03 – Washing 2	66
3.4.5	Command 04 – Draining.....	67
3.4.6	Command 05 – Filtrate Blowing 1.....	68
3.4.7	Command 06 – Filtrate Blowing 2.....	69
3.4.8	Command 07 – Core Blow 1.....	70
3.4.9	Command 08 – Core Blow 2.....	71
3.4.10	Command 9 – STOP	72
3.4.11	Command 10 – INITIAL	73
3.5	Additional Logic/Sequence	73
3.5.1	TV241 Washing 1 Control.....	73
3.6	Parameters.....	74
3.7	Configurable Parameters.....	75
3.8	Named Set	75
4	CONTROL Module Design – FR2_DISCHARGE (Discharge Logic)	76
4.1	Functional Description	76
4.1.1	Olein discharge path.....	76
4.1.2	Stearin discharge path.....	77
4.1.3	Core oil discharge path.....	78
4.1.4	Squeeze oil discharge path	78
4.1.5	Stearin tank hot water supply control.....	78
5	Module Interlocks.....	80
5.1	Functional Description	80
6	Module Force Setpoints.....	83
6.1	Functional Description	83

7	Module Permissive	92
7.1	Functional Description	92

1 EQUIPMENT MODULE CLASS DESIGN – EM_FR_CRY (CRYSTALLIZATION)

1.1 Functional Description

EM_FR_CRY is a class based equipment module that execute the crystallization sequence. EM_FR_CRY is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All required devices used in each command shall be acquired in during starting of command and they shall be released at the end of command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

1.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions (HM) and Device Failures are considered as Sentinel Monitor Conditions (SM). Only valves, motor and pump devices are considered for SM in this EM, failure monitor of Transmitters and Switches (AI modules & DI modules) will not be enabled and will not trip the sequence.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
HM02	FR2_FT2211 No Flow
HM03	Oil Level Drop Too Much, Check Leaking in Reheat Valves
HM04	Chill Water Pump Failure
HM06	Cooling Pump Failure
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Not Used
SM15	Not Used

Fail Monitor Name	Condition
SM16	Not Used
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.
SM23	Device 23 Failed.
SM24	Device 24 Failed.
SM25	Device 25 Failed.
SM26	Device 26 Failed.
SM27	Device 27 Failed.
SM28	Device 28 Failed.
SM29	Device 29 Failed.
SM30	Device 30 Failed.
SM31	Device 31 Failed.
SM32	Device 32 Failed.

Table 1: Failure Monitors

1.3

Instances

The following Table contains the module instances of the equipment module class.

*(n) denotes Crystallizer number from 01 to 10

Device Name	Module Class	Instance Name
EM	EM_FR_CRY	EM_FR2_CRY01
EM	EM_FR_CRY	EM_FR2_CRY02
EM	EM_FR_CRY	EM_FR2_CRY03
EM	EM_FR_CRY	EM_FR2_CRY04
EM	EM_FR_CRY	EM_FR2_CRY05
EM	EM_FR_CRY	EM_FR2_CRY06
EM	EM_FR_CRY	EM_FR2_CRY07
EM	EM_FR_CRY	EM_FR2_CRY08
EM	EM_FR_CRY	EM_FR2_CRY09
EM	EM_FR_CRY	EM_FR2_CRY10
DEV01	_V12_IP_130_SE	FR2_V22(n)A
DEV02	_V12_IP_130_SE	FR2_V22(n)B

Device Name	Module Class	Instance Name
DEV03	_V12_IP_130_SE	FR2_V22(n)C
DEV04	_V12_IP_130_SE	FR2_V22(n)D
DEV05	_V12_IP_130_NO	FR2_V22(n)E
DEV06	_V12_IP_130_SE	FR2_V22(n)F
DEV07	_V12_IP_130_SE	FR2_V22(n)G
DEV08	_MRSS11_130_SE	FR2_PU22(n)
DEV09	_MVS11_IL_130_SO	FR2_AG22(n)
DEV10	_CMT_130_SE	FR2_TT22(n)A
DEV11	_CMT_130_SE	FR2_TT22(n)B
DEV12	_AI_130_SE	FR2_LT22(n)
DEV13	_DI_130_SE	FR2_LSHH22(n)
DEV14	_EM_CMD8_QUEUE	FR2_FILTRATION_Q
DEV15	_EM_CMD8_QUEUE	FR2_FILLING_Q
DEV16	_FR_CRY_ICS	FR2_ICS_CRY(n)
DEV17	_AO_130_SE	FR2_TCV22(n)
DEV18	_V12_IP_130_NO	FR2_V2201
DEV19	_V12_IP_130_SE	FR2_V2211A
DEV20	_V12_IP_130_SE	FR2_V2211B
DEV21	_V12_IP_130_SE	FR2_V2211C
DEV22	_V12_IP_130_SE	FR2_V2213
DEV23	_V12_IP_130_SE	FR2_V2214
DEV24	_V12_IP_130_SE	FR2_V2215
DEV25	_V12_IP_130_SE	FR2_V2216
DEV26	_MRSS11_130_SOFT	TF1_PU91503A
DEV27	_MRSS11_130_SE	FR2_PU2211
DEV28	_CSD_130_SE	FR2_TT2211
DEV29	_O_TOT_130_SE	FR2_FT2211_TOT
DEV30	_CMTDF_130_SO	FR2_TT22(n)AB
DEV31	_MRSS11_130_SOFT	TF1_PU91503B
DEV32	_V12_IP_130_SE	FR2_REF_VLV (*tag number TBA)
DEV34	_PID_SEL_SE	FR2_TT22(n)SEL

Table 2: Instances

1.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.4.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors and Pause Timer
		A020	Reset HOLD_REQ and Hold Additional Logic
		EXIT	Common Hold Actions
T_H1000			Go to H1000
H1000	Hold Command 1	A010	Stop Pumps
		A020	Open/Close Valves
		A021	Close V22nG (cannot close until V22nE is opened)
		A030	Close Control Valves
		A035	Close Destination Valve of Reject to CR
		A036	Stop Destination Agitator of Reject to CR
		EXIT	Enable Valid Command
T_H9900			Goto H9900
H9900	Common Hold End	EXIT	Hold Logic Complete Actions
HOLD_END	Command Complete		Hold Complete

Table 3: Hold Sequence

1.4.2 Command 01 – Filling

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0009			Goto S0009
S0009	Acquire Queue Module	A009	Reset Prompt
		A010	Acquire QUEUE to proceed Filling
		EXIT	Continue Next Task
T0010			First Pass = False and no CR in Filling/Reheat/Reject then S0010
T9900A			Task Pointer = 99 And First Pass = True then S9900
T0040B			First Pass = True and Task Pointer != 99 then S0020

Step	Description	Action	Action Description
T0030			Goto S0030 if other CR in Filling/Reheat/Reject and Next to Filling is occupied
T0031			Goto S0031 if other CR in Filling/Reheat/Reject
S0010	Initialize EM	A0010	Reset Prompt
		A0020	Reset Timer (TMR1)
		EXIT	Continue Next Task
T0040			Goto S0040
S0030	Operator Prompt	EXIT	Operator Prompt: Feed Line Busy
T9901A			Goto S9901
S0031	Operator Prompt	EXIT	Operator Prompt: Next to Filling?
T9901			Goto S9901 If Next to Filling is not chosen
T0032			Goto S0032 if Next to Filling
S0032	Queue	A0010	Wait until Current CR Acquire for Filling Successfully
		EXIT	Continue Next Task
		B010	Display Queue for Filling Message
T0040C			Goto S0040
S0040	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0050			Goto S0050 if Acquire Failed
T0060			Goto S0060
T9900B			Goto S9900 if LSHH22(n) !=0 or LT22(n) !=0
S0050	Operator Prompt	EXIT	Operator Prompt (Acquisition Device Failed)
T0040A			Goto S0040
S0060	Start Filling_1	A008	Stop pumps
		A009	Close Valves
		A010	Open Valves (DEV1, 18, 20, 19/21)
		A020	Start Filling Timer
		EXIT	Continue Next Task
T0070			Goto S0070
T9900C			Goto S9900 if LSHH22(n) !=0 or LT22(n) !=0

Step	Description	Action	Action Description
S0070	Start Filling_2	A010	Open Valves (DEV1, 18, 20)
		A020	Delay 10 seconds
		A030	Start Feed Pump/Ref Valve based on Pump Selection (DEV26/31/32)
		A040	Turn TV111 to AUTO
		A050	Agitator Start & Enable FILL_AGI_SPEED & CIRCULATION_PUMP
		A060	Process Wait for DEV13 Alarm ON or DEV12 HI HI Alarm or DEV12 > Filling Stop Level or Timer reach 30 mins
		EXIT	Continue Next Task
		B010	Display Time
T0075			Goto S0075 if FT211 No Flow
T9900			Goto S9900 if 1LT22n >= FILLING Stop Level or 1LSHH22N is ON or 1LAHH22n is ON or when watchdog timer reaches 30m
S0075	Operator Prompt	EXIT	Operator Prompt for Stop Filling?
T0070A			Goto S0070 if Operator chose NO
T9902			Goto S9902 if Operator choose Yes
S9900	Complete	A010	Stop Timer
		A020	Stop Pump & Ref Valve (DEV 26, 31, 32)
		A030	Close Valves (DEV18, 19, 20, 21) and TV111
		A040	Disable All HM and SM
		A050	Release Shared Devices
		EXIT	Completed and enable next valid command
END			End of Command
S9901	Complete	A010	Release Shared Devices
		EXIT	Completed
END1			End of Command
S9902	Pause	EXIT	Command to PAUSE and Completed
END2			End of Command

Table 4: Filling Sequence

1.4.3 Command 02 – Homogenization

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False then S0010

Step	Description	Action	Action Description
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR2)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed then S0030
T0040			Goto S0040
S0030	Operator Prompt	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0020B			OAR Responded Yes, goto S0020
S0040	Homogenization	A020	Close Valves (DEV1, 2, 3, 4, 6, 7, 17)
		A030	Open Valves (DEV18, 5)
		A040	Start & Fix Agitator Speed at 45Hz and Enable CIRCULATION PUMP
		A050	Start Timer (TMR2) and wait timer to complete
		EXIT	Continue Next Task
		B010	Display Time Remaining
T9900			Goto S9900
S9900	Complete	A010	Stop Pump (DEV8) if SEMI mode
		A020	Close Valves (DEV17, 1, 2, 3) (DEV4, 5, 6, 7 if SEMI) and stop timer (TMR2)
		A030	Disable All HM and SM
		A050	Operator prompt if DEV11 < Oil Start Temp: Crystallization or Reheat?
		EXIT	Completed and set next command
END			End of Command

Table 5: Homogenization Sequence

1.4.4 Command 03 – Reheat

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands

Step	Description	Action	Action Description
T0009			Goto S0009
S0009	Acquire Queue Module	A009	Reset Prompt
		A010	Acquire Queue for REHEAT
		EXIT	Continue Next Task
T9900A			Task Pointer = 99 And First Pass = True then S9900
T0020A			First Pass = True and Task Pointer != 99 then S0020
T0010			First Pass = False & No other CR in Filling/Reheat/Reject then S0010
T0011			Goto S0011 if other CR in Filling/Reheat/Reject
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR7)
		EXIT	Continue Next Task
T0020			Goto S0020
S0011	Operator Prompt	EXIT	Operator Prompt: Feed Line Busy
T9901A			Goto S9901
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Goto S0030 if Acquire Failed
T0040			Goto S0040
S0030	Operator Prompt	EXIT	Operator Prompt: Acquire devices again?
T0020B			Goto S0020
S0040	Reheat	A009	Start Timer (TMR7), set Preheat button, record Initial Level
		A010	Stop pumps (DEV8, 26, 31, 32) and disable FILL_AGI_SPEED & CIRCULATION_PUMP
		A020	Close Valves (DEV 17,18,20,21,24,2,4,5,6,7)
		A030	Open Valves (DEV1,3,25,23,19,22)
		A040	Start and Fix Agitator (DEV9) Speed = 40
		A050	Start Pump (DEV27)
		A060	Set TV111 (DEV28) to RCAS
		EXIT	Wait for DEV11 > Oil Start Temp then Continue Next Task
		B010	Display Time

Step	Description	Action	Action Description
T9900			Goto S9900
S9900	Complete	A010	Stop Pump (DEV21,26,31,8) and close DEV32
		A020	Close Valves (DEV1-4, 6, 7, 19-25) and set DEV17,18 Output to 0
		A030	Open Valves (DEV18 & 5)
		A040	Hold Timer (TMR7)
		A050	Disable All HM and SM
		A060	Release Shared Devices
		EXIT	Completed and Proceed to Homogenization if AUTO mode
END			End of Command
S9901	Complete	A010	Release Shared Devices
		EXIT	Completed
END1			End of Command

Table 6: Reheat Sequence

1.4.5 Command 04 – Crystallization

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False then S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900B			Task Pointer = 99 And First Pass = True then S9900
S0010	Initialize EM	A010	First Pass Logic and Reset Timer (TMR3)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Devices & Start Timer	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		A030	Start/Resume Crystallization Timer (TMR3)
		EXIT	Continue Next Task
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0030			Acquire Failed
S0030	OAR Acquire Fail	A010	Acquire Failed Prompt (OK)

Step	Description	Action	Action Description
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Additional Logic	A010	Reset Cooling Segment, CCT Sequence, Auto Jump Segment
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open/Start Devices	A009	Close all other valves (DEV1-3)
		A010	Open V22nD & V22nE (DEV4 & 5)
		A011	Start Pump and Agitator (DEV8 & 9), close V22nG (DEV7)
		A020	Set TCV22n (DEV17) to CAS mode and TT22nA/B/AB (DEV10,11,30) to RCAS
		EXIT	Continue Next Task
T0300			Goto S0300
S0300	Start Cooling	A010	Start/Resume Cooling Segment, CCT sequence, Auto Jump Segment
		A020	Wait for Cooling Sequence to complete/ DEV11 < Oil Hold Temp + Compensation
		EXIT	Continue Next Task
		NS010	Set Temp SP and cooling mode
T0400			Goto S0400 if SEMI
T9900C			Goto S9900 if AUTO
S0400	SEMI mode	A010	Stop Pumps and additional logic (Auto Jump Segment, CCT Sequence, Cooling Sequence)
		A020	Close all valves (DEV1,2,3,4,6) and Open V22nE (DEV5)
		A030	Close TCV22(n) (DEV17) and V22nG (DEV7)
		EXIT	Continue Next Task
T9900A			Goto S9900
S9900	Complete	A010	Display Completion Message, Stop TMR3, Stop Auto-Jump Segment, CCT Sequence, Cooling Sequence
		A020	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 7: Crystallization Sequence

1.4.6 Command 05 – Holding

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False then S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True and then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Devices	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0300A			Task Pointer = 3 then S0300
T0200A			Task Pointer = 2 then S0200
T0100			Task Pointer = 1 then S0100
T0030			Acquire Failed
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Holding Timer (TMR4)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open/Start Devices	A009	Close all other valves (DEV1-4) & Open V22nG (DEV7)
		A010	Close V22nE (DEV5) & Open V22nF (DEV6)
		A011	Start Pump and Agitator (DEV8 & 9)
		A020	Set TCV222n (DEV17) to CAS mode and TT22nA/B/AB to RCAS (DEV10, 11 30)
		EXIT	Continue Next Task
T0201			Goto S0201
S0201	Start Holding Timer & Seq	A010	Start/Resume Holding Timer (TMR4)
		A020	Start HOLDING_SEQUENCE
		A030	Wait for Crystallizer + Holding Timer >= Min Cooling Time
		EXIT	Continue Next Task
T0300			Goto S0300

Step	Description	Action	Action Description
S0300	Stop Holding Seq & Timer	A010	Stop HOLDING_SEQUENCE
		A020	Stop Holding Timer (TMR4)
		EXIT	Continue Next Task
T0400			Goto S0400 if SEMI
T9900C			Goto S9900 if AUTO
S0400	SEMI mode	A010	Stop Pump (DEV8)
		A020	Close all valves (DEV1-4, 6) and Open V22nE (DEV5)
		A030	Close TCV22(n) (DEV17) and V22nG (DEV7)
		EXIT	Continue Next Task
T9900B			Goto S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Select Next Command
END			Command Complete

Table 8: Holding Sequence

1.4.7

Command 06 – Ready For Filtration

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or then S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True then S9900
S0010	Initialize EM	A010	First Pass Logic, Reset Timer (TMR5) and FILTRATION button
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Devices & Start Additional Logic	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		A030	Start/Resume READY FOR FILTRATION Timer (TMR5)
		A040	Start HOLDING_SEQUENCE
		EXIT	Continue Next Task
T0021			Goto S0021
T0030			Acquire Failed

Step	Description	Action	Action Description
S0021	Open/Start Devices	A009	Close all other valves (DEV1-4) & Open V22nG (DEV7)
		A010	Close V22nE (DEV5) & Open V22nF (DEV6)
		A011	Start Pump and Agitator (DEV8 & 9)
		A020	Set TCV22n (DEV17) to CAS mode and TT22nA/B/AB to RCAS (DEV10,11,30)
		EXIT	Continue Next Task
T0100			Task Pointer = 1 then S0100
T0400A			Task Pointer = 4 then S0400
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Ready for Filtration	A010	Wait for Operator to Click
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Acquire Filtration Queue	A010	Acquire QUEUE to proceed Filtration
		EXIT	Continue Next Task
T0300			Goto S0300 if no other CR in Filtration
T0400			Acquire in Queue (Go to NEXT TO FILTRATION)
T0401			Both Filtration and Next to Filtration are occupied by other CRs
S0300	Stop Timer	A010	Stop READY FOR FILTRATION Timer (TMR5)
		EXIT	Continue Next Task
T0600			Goto S0600 if SEMI
T9900B			Goto S9900 if AUTO
S0400	Next To Filtration	A010	Display Message (Current CR in Filtration and Next CR for Filtration)
		EXIT	Continue Next Task
T0300B			Goto S0300 if Filtration Queue is empty
T0500B			Goto S0500 if Operator Select Cancel Queue
S0401	Display Filtration Full Msg	A010	Display Message (Filtration and Next to Filtration are occupied)
		EXIT	Continue Next Task

Step	Description	Action	Action Description
T0500			Goto S0500
S0500	Return to Ready for Filtration	A010	Release from Queue and Return to Ready for Filtration
		EXIT	Continue Next Task
T0100B			Goto S0300
S0600	SEMI mode	A010	Stop Pump (DEV8)
		A020	Close all valves (DEV1-4, 6) and Open V22nE (DEV5)
		A030	Close TCV22(n) (DEV17) and V22nG (DEV7)
		EXIT	Continue Next Task
T9900			Goto S9900
S9900	Complete	EXIT	Completed and Set next step
END			Command Complete

Table 9: Ready for Filtration Sequence

1.4.8

Command 08 – Filtration

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or then S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True and then S9900
S0010	Initialize EM	A010	First Pass Logic and Reset Timer (TMR6)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Start Timer	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors, Start/resume timer (TMR6)
		EXIT	Continue Next Task
T0021			Task Pointer = 1 & SEMI then S0021
T0022			Task Pointer = 1 & AUTO then S0022
T0500A			Task Pointer = 5 then S0500
S0021	SEMI: Open/Start Devices	A009	Close all other valves (DEV1,3,4) & Open V22nG (DEV7)
		A010	Close V22nE (DEV5) & Open V22nF (DEV6)

Step	Description	Action	Action Description
		A011	Start Pump and Agitator (DEV8,9), Set TCV22n (DEV17) to CAS mode and TT22nA/B/AB to RCAS (DEV10,11,30)
		A020	Open V22nB (DEV2) & Start HOLDING_SEQUENCE
		EXIT	Continue Next Task
T0400			Go to S0400
S0022	Open V22nB	A010	Open V22nB (DEV2) & Start HOLDING_SEQUENCE
		EXIT	Continue Next Task
T0400A			Go to S0400
S0400	Wait for Filtration Stop Level	A030	Wait for LT22n (DEV12) reaches Filtration Stop Level
		EXIT	Continue Next Task
T0500			Go to S0500
S0500	Ending Filtration	A010	Wait for 5 seconds
		A015	Close V22nB (DEV2) and release Queue
		A020	Stop HOLDING_SEQUENCE, Pump and Agitator (DEV8,9)
		A030	Close Valve (DEV6) and Open V22nE (DEV5)
		A031	Close V22nG (DEV7) (cannot close until V22nE is opened)
		A040	Set 1TV22n (DEV17) mode to RCAS
		A060	Stop Filtration Timer (TMR6)
		EXIT	Continue Next Task
T9900			Go to S9900
S9900	Complete	A010	Release Shared Devices
		A020	Set HOLD_REQ
		EXIT	Completed and Set Batch Produced Counter
END			Command Complete

Table 10: Filtration Sequence

1.4.9

Command 09 – Reject

Step	Description	Action	Action Description
S0000	Start		Enable Valid Commands
T0010			Goto S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020

Step	Description	Action	Action Description
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR8)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Queue Module	A010	Acquire Queue for Reject
		EXIT	Continue Next Task
T0021			Goto S0021 if other CR in Filling/Reheat/Reject
T0029			Goto S0029 if no other CR in Filling/Reheat/Reject
S0021	Operator Prompt	EXIT	Operator Prompt: Feed Line Busy
T9901			Goto S9901
S0029	Skip		
T9900A			Task Pointer = 99 And First Pass = True then S9900
T0080B			Task Pointer 4 then S0080
T0070A			Task Pointer 2 then S0070
T0060A			Task Pointer 3 then S0060
T0030			Task Pointer 1 then S0030
T9902A			Task Pointer = 88 then S99002
T0240A			Task Pointer 52 then S0240
T0210B			Task Pointer 51 then S0210
S0030	Reject Prompt	EXIT	Operator Prompt: 1- With Heating, 2 - Without Heating, 3 - To CR
T0050			Goto S0050 if TK101 WITH or TK101 WITHOUT heating
T0200			Goto S0200 if Reject to Crystallizer
S0050	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0051			Acquire Failed
T0059			Goto S0059
S0051	Operator Prompt	EXIT	Operator Prompt (Acquisition Failed)
T0050A			OAR Responded Yes, goto S0050

Step	Description	Action	Action Description
S0059	Skip		
T0060			Goto S0060 if TK101 WITH heating
T0070			Goto S0070 if TK101 WITHOUT heating
S0060	With Heating	A010	TV111 (DEV28) to RCAS and enable Preheat
		A020	Open Valves (DEV1,5,19,23,24,25) and Start Agitator (DEV9) at fix speed 40
		A030	Close Valves (DEV18,20,21,22)
		A040	Delay 10 seconds
		A050	Start Pump (DEV27)
		EXIT	Continue Next Task
		B010	Display time
T0080			Goto S0080
S0070	Without Heating	A010	TV111 (DEV28) to ROUT and output to 0 and disable Preheat
		A020	Open Valves (DEV1,5,21,23,24,25) and Start Agitator (DEV9) at fix speed 40
		A030	Close Valves (DEV18,19,20,22,17)
		A040	Delay 10 seconds
		A050	Start Pump (DEV27)
		EXIT	Continue Next Task
		B010	Display Time
T0080A			Goto S0080
S0080	Process wait	EXIT	Wait for DEV12 <= Reject Oil Stop Level before Continue Next Task
		B010	Display Process Wait Message
T9900			Goto S9900 if LSHH22(n) <= Reject Oil Stop Level (%)
S0200	Select CR	A010	Operator Prompt: Enter Destination CR number
		EXIT	Continue Next Task
T0210			Goto S0210
S0210	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0220			Acquire Failed
T0230			Goto S0230

Step	Description	Action	Action Description
S0220	Operator Prompt	EXIT	Operator Prompt (Acquisition Fail)
T0210A			OAR Responded Yes, goto S0200
S0230	Check CR Level	EXIT	Display Message - Check Destination CR level
T0240			Goto S0240 if Destination CR level < Filling Stop Level
T0250			Goto S0250 if Destination CR level >= Filling Stop Level
S0240	Reject to CR	A010	Destination Valves (DEV3 & 9) mode to CAS and enable Preheat and Start TMR8
		A020	Open Valves (DEV1,19,22,23,25), open Destination valve (DEV3) and close Valves (DEV18,21,20)
		A025	Close previous Destination CR valve (DEV3) if Destination CR changed
		A030	Start Source CR Pump & Agitator (DEV9 & 27), and Destination CR Agitator (DEV9)
		A035	Stop Previous Destination CR agitator (DEV9) if Destination CR changed
		A040	TV111 (DEV28) mode to RCAS
		EXIT	Wait for Destination CR reach Stop Level then Continue Next Task
		B010	Display Process Wait message
T9902			Goto S9902
T0260			Goto S0260
S0250	Operator Prompt	EXIT	Operator Prompt: Destination CR is Full
T9903			OAR Responded OK, goto S0200
S0260	Change CRZ	EXIT	Display message when change Destination CR
T0230A			Goto S0230
S9900	Complete	A010	Stop Pump (DEV27) and Hold Timer (TMR8)
		A020	Close Valves (DEV28,1,5,27,9, 18-25)
		A030	Disable All HM and SM
		A040	Release Shared Devices
		EXIT	Completed
END			End of Command
S9901	Complete	A010	Release Shared Devices
		EXIT	Completed

Step	Description	Action	Action Description
END1			End of Command
S9902	Complete	A010	Stop Pumps (DEV27 & 8)
		A020	Close Valves (DEV1,10,19-23,25,28) and destination CR valve (DEV3) and open DEV18
		A030	Disable All HM and SM
		A040	Release Shared Devices
		EXIT	Completed
END2			End of Command

Table 11: Reject Sequence

1.4.10

Command 11 – STOP

Step	Description	Action	Action Description
S0000	Common Stop Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ, Task Pointer and First Pass
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto S0010
S0010	Stop Command 1	A010	Stop Pumps and Hold All Additional Logics/Sequence
		A020	Open/Close Valves
		A021	Close V22nG (cannot close until V22nE is opened)
		A030	Close Control Valves
		A035	Close Destination CR Valve if STOP from step Reject to CR
		A036	Stop Destination CR Agitator if STOP from step Reject to CR
		A040	Hold All Timers
		EXIT	Enable Valid Command
T9900			Go to S9900
S9900	End	A010	Release Shared Devices
		EXIT	Stop Logic Complete Actions
STOP_END			Stop Complete

Table 12: STOP sequence

1.4.11

Command 12 – INITIAL

Step	Description	Action	Action Description
S0000		A010	Disable All Hold and Sentinel Monitors

Step	Description	Action	Action Description
	Common Initial Start	A020	Reset HOLD_REQ, Task Pointer & First Pass
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto S0010
S0010	Stop Command 1	A010	Stop Pumps and Hold Additional Logics/Sequence
		A020	Open/Close Valves
		A025	Close V22nG (cannot close until V22nE is opened)
		A030	Close Control Valves
		A035	Close Destination CR Valve if STOP from step Reject to CR
		A036	Stop Destination CR Agitator if STOP from step Reject to CR
		A040	Reset All Timers
		EXIT	Enable Valid Command
T9900			Goto S9900
S9900	End	A010	Release Shared Devices
		EXIT	Initialize Logic Complete Actions
INITIAL_END			Initial Complete

Table 13: INITIAL Sequence

1.5 Additional Logic/Sequence

Additional logic or sequence are added in this Equipment Module as below:

Name	Description
FR_ICS_CRY(n)	Intelligent Cooling System
AUTOJUMP_SEGMENT	Auto-Jump Segment Sequence
CCT_SEQUENCE	Cooling Change-over Temperature Sequence (CCT)
HOLDING_SEQUENCE	Holding Sequence
FILL_AGI_SPEED	Agitator Speed for Filling
CIRCULATION_PUMP	Circulation Pump Start Sequence

Table 14: Additional Logic/Sequence for Crystallization

1.5.1 Intelligent Cooling System - FR_ICS_CRY(n)

This logic consists of 15 segments for cooling of oil which will only be activated during Crystallization sequence. Each segment has corresponding oil (FR2_TT22(n)B), water (FR2_TT22(n)A) and DT (FR2_TT22(n)AB) temperature setpoints, timer, agitator speed and mode of cooling. DT, delta temperature is the difference of oil and water temperature.

Modes of cooling include Water, Oil, DT and ADT (Active Delta T). Each segment will cool the oil based on mode selected and thus regulate control valve (FR2_TCV22(n)) to reach corresponding desired temperature setpoint.

When this sequence is started, logic will ramp the SP from previous segment SP to current segment SP within duration of timer SP. Segment will complete and proceed to next segment when desired temperature setpoint is achieved and timer SP has reached (except during DT mode, timer is not required). Agitator speed can also be set in each segment. Sequence will end when the last segment has completed.

An "Interrupt" button is used to make changes to setpoints in current running segment. It can be clicked to overwrite the current setpoint with a new setpoint key into the recipe.

A "Jog Segment" button is available to jump from the current segment to next segment when this sequence is running. It only allows jumping forward, jog backward to previous segment is not allowed.

1.5.2 Auto-Jump Segment Sequence - AUTOJUMP_SEGMENT

Sequence is active during Crystallization sequence and stop when Crystallization sequence completed.

Initial oil temperature FR2_TT22nB is recorded once sequence begins, stored in a memory. Temperature is recorded every 5 seconds. When the subsequent reading is lesser than the value in memory, the memory is replaced with the subsequent value, else the value in memory will be remained.

When the deviation of current temperature reading and value in memory is greater than Crystallization Offset (TP020), this sequence will cause the Cooling Segment (Intelligent Cooling System) to jog to the next segment. If the cooling segment is running in last segment, jogging the segment will cause the Crystallizer enter HOLDING sequence.

1.5.3 Cooling Change-over Temperature - CCT_SEQUENCE

Sequence is activated when Crystallization sequence begins. Table below shows the action triggered corresponding to each condition:

Condition	Action Triggered
DEV11 <= TP006	Open DEV6
	Start CCT Time (From Outlet), wait for timer completed
	Close DEV4
	Switch DEV17 to RCAS mode and set valve opening to TP012 for 10 secs. Return to CAS mode after timer.
DEV11 <= TP007 OR DEV6 opened longer than CCT Time (From Outlet)	Open DEV7 then close DEV5

Table 15 Cooling Change-over Temperature Sequence (CCT)

1.5.4 Holding Sequence - HOLDING_SEQUENCE

This sequence is active throughout HOLDING, READY FOR FILTRATION and FILTRATION sequences. Table below shows the action triggered corresponding to each condition:

Condition	Action Triggered
DEV11 > TP003 for more than 5s	Set DEV10 to Water mode and SP to TP005 value
DEV11 <= TP003 for more than 5s	Set DEV10 to Water mode and SP to (TP003 value – 0.2)

Table 16: Holding Sequence

1.5.5 Agitator Speed for Filling - FILL_AGI_SPEED

This logic is active, only during Filling Sequence, Step “Start Filling_2”.

Agitator speed is calculated with formula below:

$$\text{Agitator Speed (Hz)} = 30 + \frac{20}{100 - A} \times (LT - A)$$

Where,

A = Agitator start level (%)

LT = FR2_LT22(n) reading

1.5.6 Circulation Pump Start Sequence – CIRCULATION_PUMP

This sequence is active throughout FILLING & HOMOGENIZATION sequence.

Condition	Action Triggered
DEV12 PV >= TP017	Start DEV8
	Close DEV17 to 0%
DEV12 PV < TP017	Stop DEV8

Table 17: Circulation Pump Start Sequence

1.6 Parameters

The following table defines the operating, tuning parameters, report parameters and timers that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
N/A				
Tuning Parameters				
TP001	Mode: AUTO = 0, SEMI = 1	Boolean	-	0
TP002	Oil Start Temperature	Internal Reference	°C	/DEV16/START_TEMPERA
TP003	Oil Hold Temperature	Floating Point	°C	63
TP004	Oil Hold Hysteresis Temperature	Floating Point	°C	0.6
TP005	Water Holding Temperature	Floating Point	°C	23.5
TP006	CCT Inlet Temp	Floating Point	°C	47.5
TP007	CCT Outlet Temp	Floating Point	°C	47.0
TP008	Homogenization Timer	Internal Reference	Sec	60
TP009	CCT Time from Outlet	Floating Point	Sec	30
TP010	Min Cooling Time	Floating Point	Min	360
TP011	Filling Stop Level	Floating Point	%	98
TP012	TV Opening (CCT)	Floating Point	%	25

Parameter	Description	Type	Units	Default Value/Reference
TP013	Filtration Stop Level	Floating Point	%	0.5
TP014	Reject Oil Stop Level	Floating Point	%	0.5
TP015	Agitator Start Level	Floating Point	%	25
TP016	Agitator Stop Level	Floating Point	%	5
TP017	Pump Start Level	Floating Point	%	75
TP018	Destination CR	Floating Point	-	0
TP019	Filtration Button	Floating Point	-	0
TP020	Cooling/Crystallization Offset	Floating Point	°C	15
Report Parameters				
RP001	Filling Time	Floating Point	Min	0
RP002	Homogenization Time	Floating Point	Sec	0
RP003	Crystallization Time	Floating Point	Min	0
RP004	Waiting Time	Floating Point	Min	0
RP005	Ready for Filtration Time	Floating Point	Min	0
RP006	Filtration Time	Floating Point	Min	0
RP007	Reheat Time	Floating Point	Min	0
RP008	Reject Time	Floating Point	Min	0
RP009	Reheat Initial Level	Floating Point	%	0
RP010	Cooling Time Configured	Floating Point	Min	0
RP011	Service/Maintenance Indicator	Boolean	-	0
RP012	Not Used	Boolean	-	0
RP013	Totalizer before Filling	Floating Point	-	0
RP014	Totalizer after Filling	Floating Point	-	0
RP015	Batch Produced	32-bit unsigned integer	-	0
RP016	Filling Agi Final Speed	Floating Point	Hz	0

Parameter	Description	Type	Units	Default Value/Reference
Timers				
TMR1	Filling	Continuous	Min	0
TMR2	Homogenization	Time Out	Sec	60
TMR3	Crystallization	Continuous	Min	0
TMR4	Holding	Continuous	Min	0
TMR5	Ready for Filtration	Continuous	Min	0
TMR6	Filtration	Continuous	Min	0
TMR7	Reheat	Continuous	Min	0
TMR8	Reject Oil	Continuous	Min	0
TMR9	Homogenization Elapsed	Continuous	Sec	0

Table 18: Operating and Tuning Parameters

1.7 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

*(n) denotes Crystallizer number from 01 to 10

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	EM_FR2_CRY(n)
OPR_CMD_MASK	32-bit unsigned integer	4095

Table 19: Configurable Parameters

1.8 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_FR_CRY	Fractionation: Crystallization EM

Table 20: Named Set Details

Details of states for NS_FR_CRY named set are as follows,

Named Set Value	Named Set Name
00	HOLD
01	Filling
02	Homogenization

Named Set Value	Named Set Name
03	Reheat
04	Crystallizations
05	Holding
06	Ready For Filtration
08	Filtration
09	Reject
11	STOP
12	INITIAL

Table 21: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_FR_FLT (FILTRATION)

2.1 Functional Description

EM_FR_FLT is a class based equipment module that run Filtration Sequences. EM_FR_FLT is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All required devices used in each command shall be acquired in during starting of command and they shall be released at the end of command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

2.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions (HM) and Device Failures are considered as Sentinel Monitor Conditions (SM). Only valves, motor and pump devices are considered for SM in this EM, failure monitor of Transmitters and Switches (AI modules & DI modules) will not be enabled and will not trip the sequence.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
HM02	Fault Signal from Filter Press
HM03	Filtration Ready Signal Not Received
HM04	Filter Press in Manual Mode
HM05	Rope Switch Triggered
HM06	No Discharge Valve OPEN during Filtration
HM07	Olein Tank TK221 Level HI HI
HM08	Squeeze Tank Level Drop
HM09	Feeding Pressure > 4.5 barg

Fail Monitor Name	Condition
HM10	Squeezing Pressure > 16.5 barg
HM11	Squeezing Time More Than Monitoring Time
HM12	PT2212 and PT2213 Differential Pressure
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Device 15 Failed.
SM16	Device 16 Failed.
SM17	Device 17 Failed.
SM18	Device 18 Failed.
SM19	Device 19 Failed.
SM20	Device 20 Failed.
SM21	Device 21 Failed.
SM22	Device 22 Failed.
SM23	Device 23 Failed.
SM24	Device 24 Failed.
SM25	Device 25 Failed.
SM26	Device 26 Failed.
SM27	Not Used
SM28	Device 28 Failed.
SM29	Device 29 Failed.
SM30	Not Used
SM31	Not Used
SM32	Not Used

Table 22: Failure Monitor of Filtration Sequence

2.3 Instances

The following Table contains the module instances of the equipment module class.

*(n) denotes Filter Press A to B

Device Name	Module Class	Instance Name
EM	EM_FR_FLT	EM_FR2_FLT_A
EM	EM_FR_FLT	EM_FR2_FLT_B
DEV01	_V12_IP_130_SE	FR2_V22111/2
DEV02	_V12_IP_130_SE	FR2_V2213A/B
DEV03	_V12_IP_130_SE	FR2_V2214A/B
DEV04	_V12_IP_130_SE	FR2_V2221A/B
DEV05	_V12_IP_130_SE	FR2_V2212(n)A
DEV06	_V12_IP_130_SE	FR2_V2212(n)B
DEV07	_V12_IP_130_SE	FR2_V2261A/66B
DEV08	_V12_PWM_130_SO	FR2_V2217B(n)
DEV09	_V12_PWM_130_SO	FR2_V2215B(n)
DEV10	_V12_PWM_130_SO	FR2_V2216B(n)
DEV11	_V12_IP_130_SE	FR2_V2251(n)
DEV12	_V12_IP_130_SE	FR2_V2241(n)
DEV13	_V12_IP_130_SE	FR2_V2262(n)
DEV14	_V12_IP_130_SE	FR2_V2242(n)
DEV15	_V12_IP_130_SE	FR2_V2264(n)
DEV16	_MVS11_IL_130_SO	FR2_PU2211A
DEV17	_MVS11_IL_130_SO	FR2_PU2261(n)
DEV18	_CMTSL3_130_SO	FR2_PT_PU2211A
DEV19	_CMTSL_SOFT_SE	FR2_PT22612(n)
DEV21	_AI_SOFT_130_SE	FR2_PT2212(n)
DEV22	_AI_SOFT_130_SE	FR2_PT2213(n)
DEV23	_AI_130_SE	FR2_LT2241
DEV24	_DI_130_SE	FR2_LSHH2241
DEV25	_AI_130_SE	FR2_LT2261
DEV26	_DI_130_SE	FR2_LSHH2261
DEV27	_FP_STATUS	FR2_FP_(n)_STATUS
DEV28	_CMTSL3_130_SO	FR2_PT_PU2211B
DEV29	_MVS11_IL_130_SO	FR2_PU2211B

Table 23: Instances of Filtration Sequence

2.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.4.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A010	Disable All Hold and Sentinel Monitors and hold all timers
		A020	Reset HOLD_REQ, stop TK241 & TK261
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T_H1000			Goto H1000
H1000	Hold Actions	A010	Stop All Pumps
		A011	Close all Valves
		EXIT	Enable Valid Command
T_H9900			Goto H9900
H9900	Common Hold End	A010	Release Shared Devices
		EXIT	Hold Logic Complete Actions
HOLD_END			Hold Complete

Table 24: Hold Sequence

2.4.2 Command 01 – Filter Press Closing

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed

Step	Description	Action	Action Description
T0100			Task Pointer = 1 then S0100
T0400A			Task Pointer = 4 then S0400
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Start Filtration	A010	Send "Start Filtration" Signal to Andritz
		A020	Close Valves (DEV 3,7-15) Stop Pumps (DEV 16, 29, 31)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open Valves	A010	Open Valves (DEV 1,2,4,5,6)
		EXIT	Continue Next Task
T0300			Goto S0300
S0300	Close Bombdoor	A010	Send Bombdoor Close Signal (DEV27) and wait for feedback
		EXIT	Continue Next Task
T0400			Goto S0400 if SEMI
T9900C			Goto S9900 if AUTO
S0400	SEMI mode	A010	Stop or close all devices (DEV 1,2,4,5,6)
		EXIT	Continue Next Task
T9900B			Goto S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 25: Filter Press Closing Sequence

2.4.3 Command 02 – Feed Pump Start

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic & Reset Timer (TMR11)

Step	Description	Action	Action Description
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Devices	A010	Reset HOLD_REQ, stop Fill TK241 & 261, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices & Reset Fill TK241 & TK261
		EXIT	Continue Next Task
T0030			Acquire Failed
T0099			Goto S0099
T0700A			Task Pointer = 6 and SEMI then S0700
T9900			Task Pointer = 6 and AUTO then S9900
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0099	Open Valves	A010	Close Valves (DEV7-15) & Stop Pump (DEV17)
		A020	Open Valves (DEV1,2,4,5,6) and enable Fill TK241 & TK261
		EXIT	Continue Next Task
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 or 3 or 4 then S0200A
S0100	Start Pump	A010	Set Feed Pressure Control PID mode to RCAS (DEV 18/28/30)
		A020	Set Start Pressure SP (DEV18,28,30) and Start Feed Pump (DEV 16/29/31)
		A030	Send 'Sqz and Feed Step' signal to Andritz
		A040	Wait for Pressure => Start SP
		A041	Set Timer Setpoint and Description (TMR11)
		A050	Start/Resume Pressure Start Timer (TMR11) and wait for completion
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Step Pressure	A010	Set step pressure (DEV 18,28,30) and reset timer (TMR11)
		EXIT	Continue Next Task
T0201			Goto S0201

Step	Description	Action	Action Description
S0200A	Start Pump	A010	Set Feed Pressure Control PID mode to RCAS (DEV 18,28,30)
		A020	Set Start Feed Pump (DEV 16,29,31)
		EXIT	Continue Next Task
T0200C			Task Pointer = 2 or 4 Goto S0200
T0201A			Goto S0201 if TASK_PTR = 3
S0201	Wait for Pressure	A010	Wait for Pressure PV >= SP
		EXIT	Continue Next Task
T0300			If Feed Pressure < End Pressure SP, go to S0300
T0400			If Feed Pressure >= End Pressure SP, go to S0400
S0300	Start Step Timer	A009	Set Timer SP (TMR11) & Description
		A010	Start Step Timer (TMR11) and wait for completion
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0200B			Go to S0200
S0400	Start End Timer	A009	Set Timer SP (TMR11) & Description
		A010	Start End Timer (TMR11) and wait for completion
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0500			Go to S0500
S0500	Stop Feed Pump	A010	Set Feed Pressure Control PID mode to ROUT (DEV 18/28/30)
		A020	Set Feed Pressure Control PID output to 0 (DEV 18, 28, 30)
		A030	Stop Feeding Pump (DEV 16/29/31)
		A040	Close Valve V211(n) (DEV1)
		A050	Open Valve V251(n) (DEV11)
		EXIT	Continue Next Task
T0700			Goto S0700 if SEMI
T9900C			Goto S9900 if AUTO
S0700	SEMI mode	A010	Close Valves (DEV 1-6, 11)
		EXIT	Continue Next Task
T9900B			Goto S0700

Step	Description	Action	Action Description
S9900	Complete	A010	Set HOLD_REQ and Reset 'Sqz and Feed Step' Signal
		EXIT	Completed and Set Next Step
END			Command Complete

Table 26: Feed Pump Start Sequence

2.4.4

Command 03 – Adjust Sqz Pressure For Core Blow

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Open Valves	A009	Close Valves (DEV 1, 5-12 ,14) and Stop Pumps (DEV 16/29/31)
		A010	Open DEV 3 & 4
		A020	Open Valves (DEV2,13)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Start Pump	A010	Set Squeeze Pressure Control PID mode to RCAS (DEV19)
		A020	Set Core Blow Pressure (DEV19) and start Squeeze Pump (DEV17)
		A030	Wait for Pressure >= Core Pressure

Step	Description	Action	Action Description
		A040	Wait for 5 seconds
		EXIT	Continue Next Task
T0300			Goto S0300 if SEMI
T9900B			Goto S9900 if AUTO
S0300	SEMI Mode	A009	Stop Pump (DEV 17)
		A010	Close Valves (DEV 2, 3, 4, 13, 15)
		EXIT	Continue Next Task
T9900			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END	Command Complete	END	Command Complete

Table 27: Adjust Sqz Pressure For Core Blow Sequence

2.4.5

Command 04 – Squeezing

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic & Reset Timer (TMR11)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Stop Fill TK241, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices and Reset Fill TK241
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Goto S0100
T0600A			Task Pointer = 6 then S0600 if SEMI
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020

Step	Description	Action	Action Description
S0100	Open/Close Valves	A010	Close Valves (DEV 1, 5-12, 14, 15) and Stop Pumps (DEV 17, 16/ 29/ 31)
		A020	Open Valves (DEV 2, 3, 4, 13) & Enable Fill TK241
		A030	Open V251(n) (DEV11)
		EXIT	Continue Next Task
T0200			Goto S0200 if Task Pointer = 1
T0200A			Task Pointer = 3 or 4 or 2 then S0200A
S0200	Start Pump	A010	Set Squeeze Pressure Control PID mode to RCAS (DEV 19)
		A020	Set Start Pressure SP (DEV 19) and Start Squeeze Pump (DEV 17)
		A030	Send 'Sqz and Feed Step' signal to Andritz
		A040	Wait for Pressure => Start SP
		A041	Set Timer Setpoint (TMR11) and Description
		A050	Start/Resume Pressure Start Timer (TMR 11) and wait for completion
		NS01 0	Display Time Remaining to Operator
		EXIT	Completed and Set Next Step
T0300			Goto S0300
S0200A	Start Pump	A010	Set Squeeze Pressure Control PID mode to RCAS (DEV 19)
		A020	Start Squeeze Pump (DEV 17)
		EXIT	Continue Next Task
T0300C			Task Pointer 4 or 2 then S0300
T0301A			Task Pointer 3 then S0301
S0300	Step Pressure	A010	Set step pressure (DEV 19) and reset timer (TMR 11)
		EXIT	Continue Next Task
T0301			Goto S0301
S0301	Wait for Pressure	A010	Wait for Pressure PV >= SP
		EXIT	Continue Next Task
T0310			If Feed Pressure < End Pressure SP, go to S0310
T0400			If Feed Pressure >= End Pressure SP, go to S0400
S0310	Start Step Timer	A009	Set Timer SP (TMR 11) & Description
		A010	Start Step Timer (TMR11) and wait for completion

Step	Description	Action	Action Description
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0300B			Go to S0300
S0400	Start End Timer	A009	Set Timer SP (TMR11) & Description
		A010	Start End Timer and wait for completion (TMR11)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0500			Go to S0500
S0500	Stop Squeeze Pump	A010	Set Squeeze Pressure Control PID mode to ROUT (DEV 19)
		A020	Set Squeeze Pressure Control PID output to 0 (DEV 19)
		A030	Stop Squeeze Pump (DEV 17)
		EXIT	Continue Next Task
T0600			Go to S0600 if SEMI
T9900C			Goto S9900 if AUTO
S0600	SEMI mode	A010	Close Valves (DEV 2, 3 ,4, 11, 13)
		EXIT	Continue Next Task
T9900B			Goto S9900
S9900	Complete	A010	Set HOLD_REQ and reset 'Sqz and Feed Step' Signal
		EXIT	Completed and Set Next Step
END			Command Complete

Table 28: Squeezing Sequence

2.4.6

Command 05 – Core Blow 1

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task

Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 42 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Step	Description	Action	Action Description
T0020			Goto S0020
S0020	Acquire Devices	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0200			Task Pointer = 1 then S0200
T0300A			Task Pointer = 2 then S0300
T0400A			Task Pointer = 3 then S0400
S0030	Operator Prompt	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0200	Reset Timer & Close Valves	A009	Close Valves (DEV 1-4, 6, 7, 8, 10, 12, 14) & Stop Pumps (DEV 16/ 29/ 31)
		A010	Reset Core Blow Timer 1 (TMR 2)
		EXIT	Continue Next Task
T0300			Goto S0300
S0300	Open Valves & Start Timer	A010	Open Valves (DEV5, 11, 13)
		A011	Set Squeeze Pressure Control PID (DEV19) mode to RCAS
		A012	Set Core Blow Pressure (DEV 19) and start Squeeze Pump (DEV 17)
		A019	Open V215(n) (DEV9)
		A020	Start/Resume Core Blow Timer 1 and wait for completion (TMR 2)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0400			Go to S0400 if SEMI
T0500			Go to S0500 if AUTO
S0400	SEMI mode	A009	Stop Pump (DEV 17)
		A010	Close Valves (DEV 5, 9, 11, 13)
		EXIT	Continue Next Task
T9900	Go to S9900	T9900	Go to S9900
S0500	AUTO mode	A005	Close V215(n) (DEV9)
		A010	Close V212(n)A (DEV5)
		EXIT	Continue Next Task
T9900B			Go to S9900

Step	Description	Action	Action Description
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 29: Core Blow 1 Sequence

2.4.7 Command 06 – Filtrate Blowing 3

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0400A			Task Pointer = 4 then S0400
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Filtrate Blowing Timer 3 (TMR 5)
		EXIT	Continue Next Task
T0200			Go to S0200
S0200	Open Valves & Start Timer	A009	Close Valves (DEV 1,3,5,6,7,9-15) & Stop Pumps (DEV 17, 16/ 29/ 31)
		A010	Open DEV2 & 4
		A011	Open DEV8

Step	Description	Action	Action Description
		A020	Start/Resume Filtrate Blowing Timer 3 and wait for completion (TMR5)
		NS01 0	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0300			Go to S0300
S0300	Close V217(n)	A010	Close V217(n) (DEV8)
		EXIT	Continue Next Task
T0400			Go to S0400 if SEMI
T9900C			Go to S9900 if AUTO
S0400	SEMI mode	A010	Close Valves (DEV 2, 4)
		EXIT	Continue Next Task
T9900B			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 30: Filtrate Blowing 3 Sequence

2.4.8

Command 07 – Squeeze Oil

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200

Step	Description	Action	Action Description
T0300A			Task Pointer = 3 then S0300
T0400A			Task Pointer = 4 then S0400
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Timer (TMR 7)
		A020	Close Valves (DEV 1,7-15) & Stop Pumps (DEV17, 16/29/31)
		EXIT	Continue Next Task
T0200			Go to S0200
S0200	Open Valves & Start Timer	A010	Open V214(n) and V221(n) (DEV3 & 4)
		A015	Open V213(n), V212(n)A, V212(n)B (DEV2,5,6)
		A020	Start/Resume Squeeze Oil Flow Back Timer (TMR 7) and wait for completion
		NS010	Display Remaining Time
		EXIT	Continue Next Task
T0300			Go to S0300
S0300	Signal to Andritz & Close Valves	A010	Send Bombdoor Open to Andritz
		A015	Close V212(n)A, V212(n)B (DEV5,6)
		EXIT	Continue Next Task
T0400			Go to S0400 if SEMI
T9900B			Go to S9900 if AUTO
S0400	SEMI mode	A010	Close all devices (DEV 2, 3, 4)
		EXIT	Continue Next Task
T9900			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 31: Squeeze Oil Sequence

2.4.9

Command 08 – Adjust Sqz Pressure for Filtrate Blow 1

Step	Description	Action	Action Description
S0000	Start		

Step	Description	Action	Action Description
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Adjust Squeeze Timer 1 (TMR 8)
		EXIT	Continue Next Task
T0200			Go to S0200
S0200	Open Valves & Start Timer	A010	Close valves (DEV 1,5-15,) and stop pumps (DEV 17,16/29/31)
		A015	Open Valves (DEV 2,3)
		A020	Close V211(n) (DEV4)
		A030	Start/Resume Squeeze Timer 1 (TMR 8), wait for pressure <= 0.5 and timer ends
		NS010	Display Remaining Time
		EXIT	Continue Next Task
T0300			Goto S0300 if SEMI
T9900C			Goto S9900 if AUTO
S0300	SEMI mode	A010	Close Valves (DEV 2, 3)
		EXIT	Continue Next Task
T9900B			Goto S9900

Step	Description	Action	Action Description
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 32: Adjust Sqz Pressure for Filtrate Blow 1 Sequence

2.4.10

Command 09 – Cake Discharge

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Open Valves and Send Signal to Andritz	A005	Open Valves (DEV 2,3,4)
		A009	Send 'Press To Open' Signal, reset "Start Filtration" Signal and wait for FP Discharge Signal
		A010	Reset 'Press to Open' signal
		EXIT	Continue Next Task
T0200			Go to S0200
S0200		A010	Register Filtration Cycle

Step	Description	Action	Action Description
	Register Filtration Cycle	EXIT	Continue Next Task
T0300			Go to S0300
S0300	Close Valves	A010	Close Valves (DEV 2,3,4)
		EXIT	Continue Next Task
T9900B			Go to S9900
S9900	Complete	A010	Release Shared Devices if SEMI
		A020	Set HOLD_REQ
		EXIT	Completed and Set Next Step if AUTO mode
END			Command Complete

Table 33: Cake Discharge Sequence

2.4.11

Command 10 - Core Blow 2

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0150A			Task Pointer = 2 then S0150
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020

Step	Description	Action	Action Description
S0100	Reset Core Blow Timer 2	A010	Reset Core Blow Timer 2 & Reset Timer (TMR 3)
		EXIT	Continue Next Task
T0150			Go to S0150
S0150	Close/Open Valves & Start Pumps	A010	Close Valves (DEV 1-9,12,14) & Stop Pumps (DEV 16/29/31)
		A020	Open Valves (DEV 11,13)
		A030	Set Squeeze Pressure Control PID mode to RCAS (DEV 19)
		A040	Set Core Blow Pressure (DEV 19) and start Squeeze Pump (DEV 17)
		EXIT	Continue Next Task
T0200			Go to S0200
S0200	Open Valves & Start Timer	A010	Open V212(n)B (DEV2)
		A011	Open V216(n) (DEV10)
		A020	Start/Resume Core Blow Timer 2 and wait for completion (TMR3)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0300			Go to S0300
S0300	Stop Core Blow 2	A010	Close V216(n) (DEV10)
		A020	Set Squeeze Pressure Control PID mode to ROUT (DEV 19)
		A030	Set Squeeze Pressure Control PID output (DEV19) to 0 and Stop Pump (DEV17)
		A040	Close V212(n) and V251(n) (DEV6 & 11)
		A050	Close V262(n) (DEV13)
		A060	Close V264(n) (DEV15)
		EXIT	Continue Next Task
T9900			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 34: Core Blow 2 Sequence

2.4.12 Command 11 – Adjust Sqz Pressure for Filtrate Blow 2

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Adjust Squeeze Timer 2 (TMR 4)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open Valves & Start Timer	A010	Close valves (DEV 1,4-15) and stop pumps (DEV17,16/29/31)
		A015	Open Valves (DEV 2,3)
		A030	Start/Resume Squeeze Timer 1 (TMR 4, wait for pressure <= 0.5 and timer ends
		NS010	Display Remaining Time
		EXIT	Continue Next Task
T0300			Goto S0300 if SEMI
T9900C			Goto S9900 if AUTO
S0300	SEMI mode	A010	Close Valves (DEV 2, 3)
		EXIT	Continue Next Task
T9900B			Go to S9900

Step	Description	Action	Action Description
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 35: Adjust Sqz Pressure for Filtrate Blow 2 Sequence

2.4.13

Command 12 – Filtrate Blowing 4

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0201A			Task Pointer = 3 then S0201
T0300A			Task Pointer = 3 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Filtrate Blowing Timer 4 (TMR 6)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open Valves & Start Timer	A008	Open/Close Valves (DEV 1,3-7,9,10-15) & Stop Pumps (DEV17,16/29/31)
		A009	Close Valve V213(n) (DEV2)
		A010	Open V217(n) (DEV8)

Step	Description	Action	Action Description
		A020	Start/Resume Filtrate Blowing Timer 4 and wait for completion (TMR 6)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0201			Go to S0201
S0201	Close V217(n)	A010	Close V217(n) (DEV8)
		EXIT	Continue Next Task
T0300			Go to S0300 if SEMI
T9900C			Go to S9900 if AUTO
S0300	SEMI mode	A010	Close Valves (DEV 3,4)
		EXIT	Continue Next Task
T9900B			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 36: Filtrate Blowing 4 Sequence

2.4.14

Command 13 – STOP

Step	Description	Action	Action Description
S0000	Start & Acquire Devices	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ, Task Pointer & First Pass and Stop Fill TK241 & TK261
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto S0010
S0010	Stop Command	A010	Stop Pumps (DEV17, 16/29/31), Hold Additional Logics/Sequence, Set output of PID to 0 (DEV17,18/28/30)
		A020	Close all Valves (DEV1-15)
		A030	Hold All Timers
		EXIT	Enable Valid Command
T9900			Goto S9900
S9900	End	A010	Release Shared Devices
		EXIT	Hold Logic Complete Actions
STOP_END			Stop Complete

Table 37: STOP Sequence

2.4.15 Command 14 – Initial

Step	Description	Action	Action Description
S0000	Start & Acquire Devices	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ, Task Pointer & First Pass, stop Fill TK241 & TK261
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto H1000
S0010	Initial Command	A010	Stop Pumps (DEV17, 16/29/31), Hold Additional Logics/Sequence, Set output of PID to 0 (DEV17,18/28/30)
		A020	Close all Valves (DEV1-15)
		A030	Reset All Timers
		EXIT	Enable Valid Command
T9900			Goto S9900
S9900	End	EXIT	Initialize Logic Complete Actions
INITIAL_END			Initial Complete

Table 38: Initial Sequence

2.4.16 Command 15 – Filtrate Blowing 1

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed

Step	Description	Action	Action Description
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0400A			Task Pointer = 4 then S0400
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Filtrate Blowing Timer 1 (TMR 9)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open Valves & Start Timer	A009	Close Valves (DEV 1,3,5,6,7,9-15) & Stop Pumps (DEV17, 16/29/31)
		A010	Open DEV2 & 4
		A011	Open DEV8
		A020	Start/Resume Filtrate Blowing Timer 1 and wait for completion (TMR 9)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0300			Go to S0300
S0300	Close V217(n)	A010	Close V217(n) (DEV8)
		EXIT	Continue Next Task
T0400			Go to S0400 if SEMI
T9900C			Go to S9900 if AUTO
S0400	SEMI mode	A010	Close Valves (DEV 2,4)
		EXIT	Continue Next Task
T9900B			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 39: Filtrate Blowing 1 Sequence

2.4.17

Command 16 - Filtrate Blowing 2

Step	Description	Action	Action Description
S0000	Start		
T0010			First Pass = False or Restart Type != continue then S0010

Step	Description	Action	Action Description
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	First Pass Logic
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		A020	Acquire Shared Devices
		EXIT	Continue Next Task
T0030			Acquire Failed
T0100			Task Pointer = 1 then S0100
T0200A			Task Pointer = 2 then S0200
T0201A			Task Pointer = 3 then S0201
T0300A			Task Pointer = 4 then S0300
S0030	OAR Acquire Fail	A1	Acquire Failed Prompt (OK)
T0020B			OAR Responded Yes, goto S0020
S0100	Reset Timer	A010	Reset Filtrate Blowing Timer 2 (TMR 10)
		EXIT	Continue Next Task
T0200			Goto S0200
S0200	Open Valves & Start Timer	A008	Open/Close Valves (DEV 1,3-7,9-15) & Stop Pumps (DEV17, 16/29/31)
		A009	Close Valve V213(n) (DEV2)
		A010	Open V217(n) (DEV8)
		A020	Start/Resume Filtrate Blowing Timer 4 and wait for completion (TMR 10)
		NS010	Display Time Remaining to Operator
		EXIT	Continue Next Task
T0201			Go to S0201
S0201	Close V217(n)	A010	Close V217(n) (DEV8)
		EXIT	Continue Next Task
T0300			Go to S0300 if SEMI
T9900C			Go to S9900 if AUTO

Step	Description	Action	Action Description
S0300	SEMI Mode	A010	Close Valves (DEV 3,4)
		EXIT	Continue Next Task
T9900B			Go to S9900
S9900	Complete	A010	Set HOLD_REQ
		EXIT	Completed and Set Next Step
END			Command Complete

Table 40: Filtrate Blowing 2 Sequence

2.5 Additional Logic/Sequence

Additional sequences are added in this Equipment Module as below:

Name	Description
FILL_TK241_SFC	Fill TK241
FILL_TK261_SFC	Fill TK261

Table 41: Additional Sequence

2.5.1 Fill TK241

This sequence only enabled during Filtration & Squeezing Command.

Step	Description	Action	Action Description
S0000	Start		
T0100			Goto S0100 if Fill TK241 button is clicked
S0100	Open V241(n)	A010	Acquire V241(n) (DEV12)
		A020	Open V241(n) (DEV12)
T0200			Goto S0200
S0200	Close V221(n)	A010	Close V221(n) (DEV14)
			Goto S0300 if DEV23 HI HI Alm or DEV24 LSHH ON
S0300	Open V221(n)	A010	Open V221(n) (DEV4)
		A020	Close V241(n) (DEV12)
		A030	Reset Fill TK241 button
END			Command Complete, go back to S0000

Table 42: Fill TK241 Sequence

2.5.2 Fill TK261

This sequence only enabled during Filtration Command.

Step	Description	Action	Action Description
S0000	Start		
T0100			Goto S0100 if Fill TK241 button is clicked
S0100	Open V241(n)	A010	Acquire V241(n) (DEV12)
		A020	Open V241(n) (DEV12)
T0200			Goto S0200
S0200	Close V221(n)	A010	Close V221(n) (DEV14)
T0300			Goto S0300 if DEV23 HI HI Alm or DEV24 LSHH ON
S0300	Open V221(n)	A010	Open V221(n) (DEV4)
		A020	Close V241(n) (DEV12)
		A030	Reset Fill TK241 button
END			Command Complete, go back to S0000

Table 43: Fill TK261 Sequence

2.6 Parameters

The following table defines the operating, tuning parameters, report parameters and timers that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
N/A				
Tuning Parameters				
TP001	AUTO = 0, SEMI = 1	Floating Point		0
TP002	Feed Pressure Start Time (s)	Floating Point	Sec	100
TP003	Feed Pressure Step Time (s)	Floating Point	Sec	100
TP004	Feed Pressure End Time (s)	Floating Point	Sec	80

Parameter	Description	Type	Units	Default Value/Reference
TP005	Feed Start Pressure SP (barg)	Floating Point	barg	0.7
TP006	Feed Step Pressure SP (barg)	Floating Point	barg	0.7
TP007	Feed End Pressure SP (barg)	Floating Point	barg	2
TP008	Recipe Name	String	-	
TP010	Feed Speed Cut (Hz)	Floating Point	Hz	20.5
TP011	Squeeze Pressure Start Time (s)	Floating Point	Sec	200
TP012	Squeeze Pressure Step Time (s)	Floating Point	Sec	100
TP013	Squeeze Pressure End Time (s)	Floating Point	Sec	100
TP014	Squeeze Start Pressure SP (barg)	Floating Point	barg	2
TP015	Squeeze Step Pressure SP (barg)	Floating Point	barg	2.5
TP016	Squeeze End Pressure SP (barg)	Floating Point	barg	10
TP018	V214 Pressure SP (barg)	Floating Point	barg	0.23
TP019	Core Blow Pressure SP (barg)	Floating Point	barg	4
TP020	Squeezing Monitoring Timer (s)	Internal Reference	Sec	TMR1/TM_SP
TP021	Adjust Squeeze Timer 1	Internal Reference	Sec	TMR8/TM_SP
TP022	Filtrate Blow 1 Timer	Internal Reference	Sec	TMR9/TM_SP
TP023	Filtrate Blow 2 Timer	Internal Reference	Sec	TMR10/TM_SP

Parameter	Description	Type	Units	Default Value/Reference
TP026	Core Blow Timer 1 (s)	Internal Reference	Sec	TMR2/TM_SP
TP027	Core Blow Timer 2 (s)	Internal Reference	Sec	TMR3/TM_SP
TP028	Adjust Squeeze Timer 2 (s)	Internal Reference	Sec	TMR4/TM_SP
TP029	Filtrate Blow Timer 3 (s)	Internal Reference	Sec	TMR5/TM_SP
TP030	Filtrate Blow Timer 4 (s)	Internal Reference	Sec	TMR6/TM_SP
TP031	Squeeze Oil Flow Back Timer (s)	Internal Reference	Sec	TMR7/TM_SP
TP034	Fill TK241	Boolean	-	0
TP035	Fill TK261	Boolean	-	0
Report Parameters				
RP004	Filtration Cycle	Floating Point	-	0
Timers				
TMR1	Squeezing Monitoring Timer	Continuous	Sec	3600
TMR2	Core Blow 1 Timer	Time Out	Sec	60
TMR3	Core Blow 2 Timer	Time Out	Sec	60
TMR4	Adjust Squeeze Timer 2	Time Out	Sec	60
TMR5	Filtrate Blow 3 Timer	Time Out	Sec	60
TMR6	Filtrate Blow 4 Timer	Time Out	Sec	60
TMR7	Squeeze Oil Flow Back Timer	Time Out	Sec	60
TMR8	Adjust Squeeze Timer 1	Time Out	Sec	60
TMR9	Filtrate Blow 1 Timer	Time Out	Sec	60
TMR10	Filtrate Blow 2 Timer	Time Out	Sec	60
TMR11	Shared Timer	Time Out	Sec	0

Table 44: Operating and Tuning Parameters

2.7 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

*(n) denotes Filter Press A to B

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	EM_FR2_FLT_(n)
OPR_CMD_MASK	32-bit unsigned integer	131071

Table 45: Configurable Parameters

2.8 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_FR_FLT	Fractionation: Filtration EM

Table 46: Named Set Details

Details of states for NS_FR_FLT named set are as follows,

Named Set Value	Named Set Name
00	HOLD
01	Filter Press Closing
02	Feed Pump Start
03	Adjust Sqz Pressure for Core Blow
04	Squeezing
05	Core Blow 1
06	Filtrate Blowing 3
07	Squeeze Oil
08	Adjust Sqz Pressure for Filtrate Blow 1
09	Cake Discharge
10	Core Blow 2
11	Adjust Sqz Pressure for Filtrate Blow 2
12	Filtrate Blowing 4
13	STOP
14	INITIAL
15	Filtrate Blowing 1

Named Set Value	Named Set Name
16	Filtrate Blowing 2

Table 47: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_FR_WSH (WASHING)

3.1 Functional Description

EM_FR_WSH is a class based equipment module that run Washing Sequences. EM_FR_WSH is derived from the Command Driven equipment module class _EM_CMD32_130 which is defined in a separate document. Refer Functional Design Specification for the functional design requirements including faceplate and detail display. All required devices used in each command shall be acquired in during starting of command and they shall be released at the end of command. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

3.2 Failure Monitors

Conditional Interlocks to considered as Hold Monitor Conditions (HM) and Device Failures are considered as Sentinel Monitor Conditions (SM). Only valves, motor and pump devices are considered for SM in this EM, failure monitor of Transmitters and Switches (AI modules & DI modules) will not be enabled and will not trip the sequence.

Fail Monitor Name	Condition
HM01	Instance Devices Ownership Lost.
HM04	FR2_PT2211n > 1 barg
HM05	FR2_TT2241 > 70 degC
SM01	Device 1 Failed.
SM02	Device 2 Failed.
SM03	Device 3 Failed.
SM04	Device 4 Failed.
SM05	Device 5 Failed.
SM06	Device 6 Failed.
SM07	Device 7 Failed.
SM08	Device 8 Failed.
SM09	Device 9 Failed.
SM10	Device 10 Failed.
SM11	Device 11 Failed.
SM12	Device 12 Failed.
SM13	Device 13 Failed.
SM14	Device 14 Failed.
SM15	Not Used
SM16	Device 16 Failed.

Table 48: Failure Monitors

3.3 Instances

The following Table contains the module instances of the equipment module class.

*(n) denotes Filter Press number from A to B.

Device Name	Module Class	Instance Name
EM	EM_FR_WSH	EM_FR2_WSH_A
EM	EM_FR_WSH	EM_FR2_WSH_B
EM	EM_FR_WSH	EM_FR2_WSH_C
DEV01	_V12_IP_130_SE	FR2_V2213(n)
DEV02	_V12_IP_130_SE	FR2_V2214(n)
DEV03	_V12_IP_130_SE	FR2_V2212(n)A
DEV04	_V12_IP_130_SE	FR2_V2212(n)B
DEV05	_V12_PWM_130_SO	FR2_V2217B(n)
DEV06	_V12_PWM_130_SO	FR2_V2215B(n)
DEV07	_V12_PWM_130_SO	FR2_V2216B(n)

Device Name	Module Class	Instance Name
DEV08	_V12_IP_130_SE	FR2_V2251(n)
DEV09	_V12_IP_130_SE	FR2_V2242(n)
DEV10	_V12_IP_130_SE	FR2_V2241(n)
DEV11	_V12_IP_130_SE	FR2_V2264(n)
DEV12	_V12_IP_130_SE	FR2_V2243
DEV13	_CSD_130_SE	FR2_TT2241
DEV14	_MRSS11_130_SE	FR2_PU2241
DEV15	_FP_STATUS	FR2_FP_(n)_STATUS
DEV16	_AI_130_SE	FR2_PT2211(n)

Table 49: Instances

3.4 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.4.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	HOLD Start	A010	Disable All Hold and Sentinel Monitors and Pause All Timer
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T_H0020			Goto H0020
H0020	Hold	A020	Stop Pump (DEV14) and Disable WASHING1_TT241
		A030	Close Valves (DEV1-12)
		A040	Close Control Valves (DEV13)
		EXIT	Enable Valid Command
T_H9900			Goto H9900
H9900	Hold End	EXIT	Hold Logic Complete Actions
END			COMMAND COMPLETE

Table 50: Hold Sequence

3.4.2 Command 01 – Preheat

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands & Deactivate Preheat

Step	Description	Action	Action Description
T0010			First Pass = False then S0010
T0020A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030B			Acquire Failed
T0080			Goto S0080
S0030	Operator Prompt	EXIT	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0080	Check PU241/V243 Auto Mode	EXIT	Change mode of DEV12 & 14 to AUTO
T0100			Goto S0100 if DEV12 & 14 in AUTO
T0090			Goto S0090 if DEV12 & 14 not in AUTO
S0090	Operator Prompt	EXIT	Prompt DEV12 & 14 not in AUTO mode
T9900D			Goto S9900 when OAR responded OK
S0100	Wait V2242 Close	EXIT	Wait V242 to close (DEV9)
T0110			Goto S0110 if DEV9 OPEN
T0120			Goto S0120 if DEV9 CLOSE
S0110	Operator Prompt	A010	Operator Prompt: DEV9 is OPEN
T9900E			Goto S9900 when OAR responded OK
S0120	Start Preheat	A010	Open Valves (DEV 12)
		A020	Start Pump (DEV 14)
		A030	Turn TT241 (DEV13) Auto
		EXIT	Continue Next Task
T0130			Goto S0130 if DEV13 SP >= PV

Step	Description	Action	Action Description
T0130A			Goto S0130 if Preheat Deactivation
S0130	Delay 30 seconds	EXIT	Wait for 30 secs and Continue Next Task
		B010	Display remaining time
T9900			Goto S9900
S9900	Complete	A010	Stop Pump (DEV14)
		A020	Close Valves (DEV12 & 13)
		A030	Disable All HM and SM
		EXIT	Completed and Set Next Step if Operator click START
END			Command Complete

Table 51: Preheat Sequence

3.4.3

Command 02 – Washing 1

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0015A			First Pass = True and Task Pointer != 99 then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR1)
		EXIT	Continue Next Task
T0015			Goto S0015
S0015	Check TT241	EXIT	Check Temperature (DEV13)
T0020			Goto S0020 if TT241 PV > SP
T9901			Goto S9901 if TT241 PV <= SP
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Goto S0030 if Acquire Failed
T0040			Goto S0040
S0030	Operator Prompt	EXIT	Operator Prompt (Acquisition Failed)
T0020B			Goto S0020 when OAR responded OK

Step	Description	Action	Action Description
S0040	Start Washing 1	A020	Close Valves (DEV 2,5,6,7,8,11)
		A030	Open Valves (DEV 1,3,4,9,10)
		A040	Close Valve (DEV12)
		A050	Run Timer (TMR1) & enable WASHING1_TT241
		EXIT	Continue Next Task if Timer Ends
		B010	Display Remaining Time
T9900			Goto S9900
S9900	Complete	A010	Stop Pump (DEV14) if SEMI, hold timer (TMR1) and disable WASHING1_TT241
		A020	Close Valves (DEV1,3,4,9,10) if SEMI
		A030	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete
S9901	Preheat	EXIT	Completed and start Preheat command
END2			Command Complete

Table 52: Washing 1 Sequence

3.4.4

Command 03 – Washing 2

Step	Description	Action	Action Description
S0000	Start	EXIT	Start
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR2)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed
T0040			Goto S0040

Step	Description	Action	Action Description
S0030	Operator Prompt	A010	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040	Start Washing 2	A020	Close Valves (DEV 5,6,7,8,11,12)
		A030	Open Valves (DEV 1,2,3,4,9,10) and enable WASHING1 TT241
		A040	Start Pump (DEV 14)
		A050	Run Timer (TMR 2) and wait for completion
		EXIT	Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Stop Pump (DEV 14) and disable WASHING1 TT241
		A020	Close DEV 1,2,10 if SEMI and close Valves (DEV3,4,9)
		A030	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete

Table 53: Washing 2 Sequence

3.4.5

Command 04 – Draining

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR3)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed
T0040			Goto S0040

Step	Description	Action	Action Description
S0030	Operator Prompt	EXIT	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040	Start Draining	A010	Stop Pump (DEV 14)
		A020	Close Valves (DEV3-9,13,11,12)
		A030	Run Timer (TMR 3)
		A040	Open Valves (DEV 1,2,10)
		EXIT	Wait for completion of timer then Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Close Valves (DEV1 & 10) if SEMI, close DEV2 and hold timer (TMR3)
		A020	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete

Table 54: Draining Sequence

3.4.6 Command 05 – Filtrate Blowing 1

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR4)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Devices	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
S0030	Operator Prompt	A010	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040		A010	Stop Pump (DEV 14)

Step	Description	Action	Action Description
	Start Filtrate Blowing 1	A020	Close Valves (DEV 2-4, 6-9,11-13)
		A030	Open Valves (DEV 1,10)
		A040	Open Andritz Valve (DEV 5)
		A050	Run Timer (TMR 4) and wait for completion
		EXIT	Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Close DEV10 if SEMI, close Valves (DEV 1,5) and hold TMR4
		A020	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete

Table 55: Filtrate Blowing 1

3.4.7

Command 06 – Filtrate Blowing 2

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR5)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed
T0040			Goto S0040
S0030	Operator Prompt	A010	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040	Start Filtrate Blowing 2	A010	Stop Pump (DEV 14)
		A020	Close Valves (DEV1,3,4,6-9,11-13)

Step	Description	Action	Action Description
		A030	Open Valves (DEV 2,10)
		A040	Open Andritz Valve (DEV 5)
		A050	Start Timer (TMR 5) and wait for completion
		EXIT	Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Close Valves (DEV 2,5,10) and hold timer (TMR5)
		A020	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete

Table 56: Filtrate Blowing 2

3.4.8

Command 07 – Core Blow 1

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR6)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed
T0040			Goto S0040
S0030	Operator Prompt	A010	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040	Start Core Blow 1	A010	Stop Pump (DEV 14)
		A020	Close Valves (DEV5,7,1,2,4,9-13)

Step	Description	Action	Action Description
		A030	Open Valves (DEV 3,8)
		A040	Open Andritz Valve (DEV 6)
		A050	Run Timer (TMR 6) and wait for completion
		EXIT	Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Close DEV8 if SEMI, close Valves (DEV 3 & 6) and hold timer TMR6
		A020	Disable All HM and SM
		EXIT	Completed and Set Next Step
END			Command Complete

Table 57: Core Blow 1 Sequence

3.4.9

Command 08 – Core Blow 2

Step	Description	Action	Action Description
S0000	Start	EXIT	Enable Valid Commands
T0010			First Pass = False or Restart Type != continue then S0010
T0020A			First Pass = True and Task Pointer != 99 and Restart Type = continue then S0020
T9900A			Task Pointer = 99 And First Pass = True and Restart Type = continue then S9900
S0010	Initialize EM	A010	Reset Prompt
		A020	Reset Timer (TMR7)
		EXIT	Continue Next Task
T0020			Goto S0020
S0020	Acquire Device	A010	Reset HOLD_REQ, Enable Required Hold and Sentinel Monitors
		EXIT	Acquire Devices
T0030			Acquire Failed
T0040			Goto S0040
S0030	Operator Prompt	A010	Operator Prompt (Acquisition Failed)
T0020B			OAR Responded Yes, goto S0020
S0040	Start Core Blow 2	A010	Stop Pump (DEV 14)
		A020	Close Valves (DEV 5,6,13,1-3,9-12)

Step	Description	Action	Action Description
		A030	Open Valves (DEV 4,8)
		A040	Open Andritz Valve (DEV 7)
		A050	Start Timer (TMR 7) and wait for completion
		EXIT	Continue Next Task
		B010	Display Timer
T9900			Goto S9900
S9900	Complete	A010	Close Valves (DEV 4,7) and hold timer (TMR7)
		A015	Close V251(n) (DEV8)
		A020	Disable All HM and SM
		A040	Send to Andritz (DEV 15)
		A050	Receive from Andritz
		A060	Reset After Washing Signal (DEV 15) and register Washing Cycle
		A070	Release Shared Devices
		EXIT	Completed
END			Command Complete

Table 58: Core Blow 2 Sequence

3.4.10

Command 9 – STOP

Step	Description	Action	Action Description
S0000	STOP Start	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto S0010
S0010	Stop Command	A010	Reset all Timers
		A020	Stop Pump (DEV14) and disable WASHING1 TT241
		A030	Close Valves (DEV1-12)
		A040	Close Control Valves (DEV13)
		EXIT	Enable Valid Command
T9900			Go to S9900
S9900	End	A010	Release Shared Devices
		EXIT	Stop Logic Complete Actions
STOP_END			Stop Complete

Table 59: STOP Sequence

3.4.11 Command 10 – INITIAL

Step	Description	Action	Action Description
S0000	Acquire Shared Devices	A010	Disable All Hold and Sentinel Monitors
		A020	Reset HOLD_REQ
		A030	Acquire Shared Devices
		EXIT	Common Hold Actions
T0010			Goto S0010
S0010	Initial Command	A010	Reset all Timers
		A020	Stop Pump (DEV14) and disable WASHING1_TT241
		A030	Close Valves (DEV1-12)
		A040	Close Control Valves (DEV13)
		EXIT	Enable Valid Command
T9900			Goto S9900
S9900	End	EXIT	Initial Logic Complete Actions
INITIAL_END			Initial Complete

Table 60: INITIAL Sequence

3.5 Additional Logic/Sequence

Additional logic is added in this Equipment Module as below:

Name	Description
WASHING1_TT241	TV241 Washing 1 Control

Table 61: Additional Logic

3.5.1 TV241 Washing 1 Control

This logic is only enabled during Washing 1 command.

Condition	Action Triggered
DEV13 PV > 67 DegC for 10s	Turn DEV13 to ROUT, set output to 0%
	Start DEV14
DEV13 PV > 67 DegC for another 60s	Turn DEV13 to ROUT, set output to 0%
	Stop DEV14
DEV13 PV <= 67 DegC	Turn DEV13 to AUTO
	Start DEV14

Table 62: TV241 Washing 1 Control

3.6 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
N/A				
Tuning Parameters				
TP001	Mode: AUTO = 0, SEMI = 1	Boolean	-	0
TP002	Washing 1 Timer	Internal References	Sec	TMR1/TM_SP
TP003	Washing 2 Timer	Internal References	Sec	TMR2/TM_SP
TP004	Draining Timer	Internal References	Sec	TMR3/TM_SP
TP005	Filtrate Blowing 1 Timer	Internal References	Sec	TMR4/TM_SP
TP006	Filtrate Blowing 2 Timer	Internal References	Sec	TMR5/TM_SP
TP007	Core Blow 1 Timer	Internal References	Sec	TMR6/TM_SP
TP008	Core Blow 2 Timer	Internal References	Sec	TMR7/TM_SP
Report Parameters				
RP001	Washing Cycle	32-bit unsigned integer	-	0
RP002	Preheat Deactivation	8-bit unsigned integer	-	0
Timers				
TMR1	Washing 1	Timeout	Sec	10
TMR2	Washing 2	Timeout	Sec	10
TMR3	Draining	Timeout	Sec	10
TMR4	Filtrate Blowing 1	Timeout	Sec	10
TMR5	Filtrate Blowing 2	Timeout	Sec	10
TMR6	Core Blow 1	Timeout	Sec	10
TMR7	Core Blow 2	Timeout	Sec	10

Table 63: Parameters of Washing Sequence

3.7 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

*(n) denotes Filter Press A to B

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	EM_FR2_WSH_(n)
OPR_CMD_MASK	32-bit unsigned integer	2047

Table 64: Configurable Parameters

3.8 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_FR_WSH	Fractionation: Washing EM

Table 65: Named Set Details

Details of states for NS_FR_WSH named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Preheat
02	Washing 1
03	Washing 2
04	Draining
05	Filtrate Blowing 1
06	Filtrate Blowing 2
07	Core Blow 1
08	Core Blow 2
09	STOP
10	INITIAL

Table 66: Name Set State Details

4 CONTROL MODULE DESIGN – FR2_DISCHARGE (DISCHARGE LOGIC)

4.1 Functional Description

The Discharge logic is configured using combination of customised classless control module named FR2_DISCHARGE and module force setpoint and interlocks.

Discharge logic consists of:

- Olein discharge path
- Stearin discharge path
- Core oil discharge path
- Squeeze oil discharge path
- Stearin tank hot water supply control

4.1.1 Olein discharge path

Devices controlled by this logic as below:

Module Name
FR2_PU2221
FR2_PU2223
FR2_V2222A
FR2_V2222B
FR2_V2223A
FR2_V2223B
FR2_V2224A
FR2_V2224B
FR2_V2224C

Table 67: Devices used for Olein Discharge Path

Actions of each valves and pumps based on discharge path selection is configured as module interlocks and force setpoints. Refer to Section 5 and 6 for all interlocks and force setpoints configured in devices above.

Parameter OLEIN_DISC in module FR2_DISCHARGE is the input from graphic to select the discharge path of olein as below:

Parameter	Value	Description
OLEIN_DISC	0	Hot Shock
	1	Tank Farm
	2	2HM211A
	3	2HM211B

Parameter	Value	Description
	5	STOP

Table 68: Configurable Parameters of Olein Discharge

Below shows the additional logic implemented from customised module FR2_DISCHARGE.

Condition	Action Triggered
Initial Button is clicked	Turn all devices to CAS mode
	Stop or Close all devices

Table 69: Additional Logic for Olein Discharge Path

4.1.2

Stearin discharge path

Devices controlled by this logic as below:

Module Name
FR2_PU2231A
FR2_V2231A
FR2_V2233
FR2_PU2231B
FR2_V2231B

Table 70: Devices used for Stearin Discharge Path

Actions of each valves and pumps based on discharge path selection is configured as module interlocks and force setpoints. Refer to Section 5 and 6 for all interlocks and force setpoints configured in devices above.

Parameters below in module FR2_DISCHARGE are the input from graphic to select the discharge path of stearin and start/stop:

*(n) denotes Stearin Tank A to C

Parameter	Value	Description
STEARIN_DISC_(n)	0	Transfer
	1	Circulation
START_S_(n)	0	Stop
	1	Start
V233N_TIMER	Settable	User settable timer for V233N to open

Table 71: Configurable Parameters of Stearin Discharge

Below shows the additional logic implemented from customised module FR2_DISCHARGE.

Condition	Action Triggered
Stearin Tank (n) Initial Button is clicked	Turn all devices (n) to CAS mode

Condition	Action Triggered
FR2_V2233 is Closed	Open FR2_V2233N if mode = CAS
FR2_V2233N opened for V233N_TIMER duration	Close FR2_V2233N if mode = CAS

Table 72: Additional Logic for Stearin Discharge Path

4.1.3 Core oil discharge path

Devices controlled by this logic as below:

Module Name
FR2_PU2251
FR2_V2252
FR2_V2253

Table 73: Devices used for Core Oil Discharge Path

Actions of each valves and pumps based on discharge path selection is configured as module interlocks and force setpoints. Refer to Section 5 and 6 for all interlocks and force setpoints configured in devices above.

Parameters below in module FR2_DISCHARGE are the input from graphic to select the discharge path of core oil and start/stop:

Parameter	Value	Description
COREOIL_DISC	0	Circulate
	1	Transfer
	2	STOP

Table 74: Configurable Parameters of Core Oil Discharge

Below shows the additional logic implemented from customised module FR2_DISCHARGE.

Condition	Action Triggered
FR2_LSL2251 is ON	Set COREOIL_DISC = 2
FR2_LSHH2251 is ON	Set COREOIL_DISC = 1

Table 75: Additional Logic for Core Oil Discharge Path

4.1.4 Squeeze oil discharge path

The discharge is controlled by operator, only interlocks are configured for FR2_PU2261(n), FR2_V2265(n) and FR2_V2262(n). Refer to Section 5 and 6 for the interlocks configured.

4.1.5 Stearin tank hot water supply control

Devices controlled by this logic as below:

Module Name
FR2_PU2230
FR2_V2232A
FR2_V2232B
FR2_V2230
FR2_TT2230

Table 76: Devices used for Stearin tank hot water supply control

Actions of each valves and pumps for stearin hot water supply control is configured as module interlocks and force setpoints. Refer to Section 5 and 6 for all interlocks and force setpoints configured in devices above.

Below shows the additional logic implemented from customised module FR2_DISCHARGE.

Condition	Action Triggered
Initial Button is clicked	Turn all devices to CAS mode
	Stop or Close all devices
Stop Button is clicked	Turn all devices to CAS mode
	Stop or Close all devices

Table 77: Additional Logic for Stearin tank hot water supply control

5 MODULE INTERLOCKS

5.1 Functional Description

The section below list down the interlock configurations.

*(n) denotes Crystallizer number from 01 to 10

*(x) denotes character A to C

Module	Name	Description	Expression	Interlock State	Delay Off (s)	Delay On (s)
FR2_AG22(n)	Condition 2	LT22(n) <= Agitator Stop Level	//FR2_LT22(n)/PV.CV' <= '//EM_FR2_CRY(n)/TP016_VALUE.CV'	STOP	NO	NO
FR2_PU2211(x)	Condition 2	PSH2211(x) is ON	'//FR2_PSH2211(x)/DI1/DISC_ACT.CV' = 1	STOP	NO	NO
FR2_PU2211(x)	Condition 3	TSH2211(x) is ON	'//FR2_TSH2211(x)/DI1/DISC_ACT.CV' = 1	STOP	NO	NO
FR2_PU2221	Condition 2	LSHH2223 is ON	'//FR2_LSHH2223/PV_D.CV' = 1	STOP	NO	NO
FR2_PU2221	Condition 3	LT2223 HI HI	'//FR2_LT2223/HI_HI_ALM.CUALM'	STOP	NO	NO
FR2_PU2223	Condition 2	LSLL2223 is ON	'//FR2_LSLL2223/PV_D.CV' = 1	STOP	NO	NO
FR2_PU2223	Condition 3	LT2223 < LO	'//FR2_LT2223/LO_ALM.CUALM' != 0	STOP	NO	NO
FR2_PU2223	Condition 4	LT2223 < LO LO	'//FR2_LT2223/LO_LO_ALM.CUALM' != 0	STOP	NO	NO
FR2_PU2223	Condition 5	LT2223 > HI	'//FR2_LT2223/HI_ALM.CUALM' != 0	START	NO	NO
FR2_PU2230	Condition 2	LALL1230 is ON	'//FR2_LT2230/AI1/LO_LO_ACT.CV' != 0	STOP	NO	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

PT. Wilmar Nabati Indonesia - DCS Serang Project

Revision 0 – 2019-Aug-16 – Page 81 of 92

3258585

Module	Name	Description	Expression	Interlock State	Delay Off (s)	Delay On (s)
FR2_PU2241	Condition 2	LT2241 LO LO Alarm	'//FR2_LT2241/LO_LO_ALM.CUALM' > 0	STOP	NO	NO
FR2_PU2261(x)	Condition 2	LALL1261 ON	'//FR2_LT2261/AI1/LO_LO_ACT.CV' != 0	STOP	NO	NO
FR2_PU2261(x)	Condition 3	PSH1261(x) is ON	'//FR2_PSH1261(x)/DI1/DISC_ACT.CV' = 1	STOP	NO	NO
FR2_PU2261(x)	Condition 4	TSH1261(x) is ON	'//FR2_TSH1261(x)/DI1/DISC_ACT.CV' = 1	STOP	NO	NO
FR2_PU1401C	Condition 2	Level Indicator Control Alarm Low LICA-1401	'//FR2_LT1401/LO_ALM.CUALM' = TRUE	STOP	NO	NO
FR2_PU1401C	Condition 3	Level Switch Alarm Low Low LALL-1401	'//FR2_LSLL1401/PV_D' = 1	STOP	NO	NO
FR2_PU1401D	Condition 2	Level Indicator Control Alarm Low LICA-1401	'//FR2_LT1401/LO_ALM.CUALM' = TRUE	STOP	NO	NO
FR2_PU1401D	Condition 3	Level Switch Alarm Low Low LALL-1401	'//FR2_LSLL1401/PV_D' = 1	STOP	NO	NO
FR2_PU1403A	Condition 2	Level Indicator Control Alarm Low LICA-1402	'//FR2_LT1402/LO_ALM.CUALM' = TRUE	STOP	NO	NO
FR2_PU1403A	Condition 3	Level Switch Alarm Low Low LALL-1402	'//FR2_LSLL1402/PV_D' = 1	STOP	NO	NO
FR2_PU1403B	Condition 2	Level Indicator Control Alarm Low LICA-1402	'//FR2_LT1402/LO_ALM.CUALM' = TRUE	STOP	NO	NO
FR2_PU1403B	Condition 3	Level Switch Alarm Low Low LALL-1402	'//FR2_LSLL1402/PV_D' = 1	STOP	NO	NO
FR2_PU3012C	Condition 2	Level Switch Alarm Low Low LALL-3012A	'//FR2_LSLL3012A/PV_D' = 1	STOP	NO	NO
FR2_PU3012D	Condition 2	Level Switch Alarm Low Low LALL-3012A	'//FR2_LSLL3012A/PV_D' = 1	STOP	NO	NO
FR2_PU302A	Condition 2	Level Switch Alarm Low Low LALL-301	'//FR2_LSLL301/PV_D' = 1	STOP	NO	NO

Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

PT. Wilmar Nabati Indonesia - DCS Serang Project

Revision 0 – 2019-Aug-16 – Page 82 of 92

3258585

Module	Name	Description	Expression	Interlock State	Delay Off (s)	Delay On (s)
FR2_PU302B	Condition 2	Level Switch Alarm Low Low LALL-301	'//FR2_LSLL301/PV_D' = 1	STOP	NO	NO
FR2_PU602B	Condition 2	Level Switch Alarm High LCA-602	'//FR2_LSH602/PV_D' = 1	START	NO	NO
FR2_PU902B	Condition 2	Level Control High Alarm LCA-902	'//FR2_LSH902/PV_D' = 1	START	NO	NO
FR2_V22(n)E	Condition 1	FR2_V22(n)G is CLOSED	'//FR2_V22(n)G/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	CLOSE	NO	NO
FR2_V22(n)G	Condition 1	FR2_V22(n)E is closed	'//FR2_V22(n)E/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO	NO
FR2_V2262(x)	Condition 1	FR2_V2265(x) is OPEN	'//FR2_V2265(x)/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO	NO
FR2_V2265(x)	Condition 1	FR2_V2262(x) is OPEN	'//FR2_V2262(x)/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO	NO
FR2_TT22(n)AB	Condition 1	TT22(n)A <= Seg Wtr SP (DT)	('//FR2_ICS_CRY(n)/START.CV' = 1) AND rem ICS is active ('//FR2_ICS_CRY(n)/COOLING_SEQ/MODE_PV.CV' = 3) AND rem in DT mode ('//FR2_TT22(n)A/PV.CV' <= '//FR2_ICS_CRY(n)/COOLING_SEQ/WTR_RECIP E_PV.CV')	0%	NO	NO
FR2_TT22(n)B	Condition 1	TT22(n)A <= Seg Wtr SP (Oil)	('//FR2_ICS_CRY(n)/START.CV' = 1) AND rem ICS is active ('//FR2_ICS_CRY(n)/COOLING_SEQ/MODE_PV.CV' = 1) AND rem in Oil mode ('//FR2_TT22(n)A/PV.CV' <= '//FR2_ICS_CRY(n)/COOLING_SEQ/WTR_RECIP E_PV.CV')	0%	NO	NO

Table 78: Module Interlocks


6 MODULE FORCE SETPOINTS

6.1 Functional Description

The section below list down the force setpoints configurations.

*(n) denotes Crystallizer number from 01 to 22

*(x) denotes character A to C

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_PU2221	Condition 1	FR2_LSL2221 ON	'//FR2_LSL2221/DI1/DISC_ACT.CV' = 1	STOP	NO
FR2_PU2221	Condition 2	FR2_LSH2221 ON	('//FR2_LSH2221/DI1/DISC_ACT.CV' = 1) AND ('//FR2_V2222A/PV_D.CV' = ' _EDC_VLV_PV:OPEN' OR '//FR2_V2222B/PV_D.CV' = ' _EDC_VLV_PV:OPEN' OR '//FR2_V2223A/PV_D.CV' = ' _EDC_VLV_PV:OPEN' OR '//FR2_V2223B/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	START	NO
FR2_PU2221	Condition 3	STOP is clicked	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 5)	STOP	NO
FR2_PU2230	Condition 1	V2232A is OPEN	('//FR2_V2232A/PV_D.CV' = ' _EDC_VLV_PV:OPEN') AND ('//FR2_DISCHARGE/START.CV' = 1)	START	NO
FR2_PU2230	Condition 2	V2232B is OPEN	('//FR2_V2232B/PV_D.CV' = ' _EDC_VLV_PV:OPEN') AND ('//FR2_DISCHARGE/START.CV' = 1)	START	NO
FR2_PU2230	Condition 4	V2232A/B are CLOSE	('//FR2_V2232A/PV_D.CV' = ' _EDC_VLV_PV:CLOSED') AND ('//FR2_V2232B/PV_D.CV' = ' _EDC_VLV_PV:CLOSED')	STOP	NO
FR2_PU2231(x)	Condition 1	LSL2231(x) ON	//FR2_LSL2231A/DI1/DISC_ACT.CV' = 1	STOP	NO
FR2_PU2231(x)	Condition 2	LSH2231(x) ON & CIRCULATE	('//FR2_LSH2231(x)/DI1/DISC_ACT.CV' = 1) AND ('//FR2_DISCHARGE/STEARIN_DISC_(x).CV' = 1) AND ('//FR2_V2231(x)/PV_D.CV' = ' _EDC_VLV_PV:OPEN') AND ('//FR2_DISCHARGE/START_S_(x).CV' = 1)	START	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 84 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_PU2231(x)	Condition 3	LSLL2231 ON	'//FR2_LSLL2231/DI1/DISC_ACT.CV' = 1	STOP	NO
FR2_PU2231(x)	Condition 4	LSH2231(x) ON & TRANSFER	('//FR2_LSH2231(x)/DI1/DISC_ACT.CV' = 1) AND ('//FR2_DISCHARGE/STEARIN_DISC_(x).CV' = 0) AND ('//FR2_V2233/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND ('//FR2_DISCHARGE/START_S_(x).CV' = 1)	START	NO
FR2_PU2231(x)	Condition 5	STOP is clicked	('//FR2_DISCHARGE/START_S_(x).CV' = 0)	STOP	NO
FR2_PU2241	Condition 1	LSHH2241 OFF for 60s	'//FR2_LSHH2241/DI1/DISC_ACT.CV' = 0	STOP	60
FR2_PU2241	Condition 2	LSHH2241 ON & V244 OPEN	('//FR2_LSHH2241/DISC_ALM.CUALM') AND ('//FR2_V2244/PV_D.CV' = 'EDC_VLV_PV:OPEN')	START	NO
FR2_PU2251	Condition 1	LSH/LSHH2251 ON & TRANSFER	((('//FR2_LSH2251/DI1/DISC_ACT.CV' = 1) OR ('//FR2_LSHH2251/DI1/DISC_ACT.CV' = 1)) AND ('//FR2_DISCHARGE/COREOIL_DISC.CV' = 1) AND ('//FR2_V2253/PV_D.CV' = 'EDC_VLV_SP:OPEN'))	START	NO
FR2_PU2251	Condition 2	LSH/LSHH2251 ON & CIRCULATE	('//FR2_LSH2251/DI1/DISC_ACT.CV' = 1) AND ('//FR2_DISCHARGE/COREOIL_DISC.CV' = 0) AND ('//FR2_V2252/PV_D.CV' = 'EDC_VLV_SP:OPEN')	START	NO
FR2_PU2251	Condition 3	STOP is clicked	'//FR2_DISCHARGE/COREOIL_DISC.CV' = 2	STOP	NO
FR2_PU1403A	Condition 1	One of V22nF OPEN	('//FR2_V2201F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2202F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2203F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2204F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2205F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2206F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2207F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2208F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2209F/PV_D.CV' = 'EDC_VLV_PV:OPEN') OR ('//FR2_V2210F/PV_D.CV' = 'EDC_VLV_PV:OPEN')	START	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 85 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_PU1403A	Condition 2	None of V22nF is OPEN	('//FR2_V2201F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2202F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2203F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2204F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2205F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2206F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2207F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2208F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2209F/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2210F/PV_D.CV' != ' _EDC_VLV_PV:OPEN')	STOP	NO
FR2_PU301A	Condition 1	One of V22nD OPEN	('//FR2_V2201D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2202D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2203D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2204D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2205D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2206D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2207D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2208D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2209D/PV_D.CV' = ' _EDC_VLV_PV:OPEN') OR ('//FR2_V2210D/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	START	NO
FR2_PU301A	Condition 2	None of V22nD is OPEN	('//FR2_V2201D/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2202D/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2203D/PV_D.CV' != ' _EDC_VLV_PV:OPEN') AND ('//FR2_V2204D/PV_D.CV' != ' _EDC_VLV_PV:OPEN')	STOP	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 86 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
			AND ('//FR2_V2205D/PV_D.CV' =! '_EDC_VLV_PV:OPEN') AND ('//FR2_V2206D/PV_D.CV' =!'_EDC_VLV_PV:OPEN') AND ('//FR2_V2207D/PV_D.CV' =!'_EDC_VLV_PV:OPEN') AND ('//FR2_V2208D/PV_D.CV' =! '_EDC_VLV_PV:OPEN') AND ('//FR2_V2209D/PV_D.CV' =!'_EDC_VLV_PV:OPEN') AND ('//FR2_V2210D/PV_D.CV' =!'_EDC_VLV_PV:OPEN')		
FR2_REF_VLV	Condition 1	Switch from Pump A	('//TF1_PU91503A/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') AND ('//TF1_PU91503A/TP01.CV' = 2)	OPEN	NO
FR2_REF_VLV	Condition 2	Switch from Pump B	('//TF1_PU91503B/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') AND ('//TF1_PU91503A/TP01.CV' = 2)	OPEN	NO
FR2_REF_VLV	Condition 3	Switch to Pump A	('//TF1_PU91503A/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') AND ('//TF1_PU91503A/TP01.CV' = 0)	CLOSE	NO
FR2_REF_VLV	Condition 4	Switch to Pump B	('//TF1_PU91503B/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') AND ('//TF1_PU91503A/TP01.CV' = 1)	CLOSE	NO
FR2_V2214A	Condition 1	PT22111A > V12141 Set Pressure	('//FR2_PT2211A/PV.CV' > '//EM_FR2_FLT_A/TP018_VALUE.CV') AND (('//EM_FR2_FLT_A/A_TARGET.CV' = 2) AND ('//EM_FR2_FLT_A/A_STATE.CV' = 50)) AND (('//EM_FR2_FLT_A/PUMP_SEL.CV' = 1 AND '//FR2_PU2211A/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') OR ('//EM_FR2_FLT_A/PUMP_SEL.CV' = 2 AND '//FR2_PU2211B/PV_D.CV' = '_EDC_MTR2_PV:RUNNING'))	OPEN	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 87 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
			REM active during feed pump start		
FR2_V2214B	Condition 1	PT22111B > V12142 Set Pressure	('//FR2_PT2211B/PV.CV' > '//EM_FR2_FLT_B/TP018_VALUE.CV') AND (('//EM_FR2_FLT_B/A_TARGET.CV' = 2) AND ('//EM_FR2_FLT_B/A_STATE.CV' = 50)) AND (('//EM_FR2_FLT_B/PUMP_SEL.CV' = 1 AND '//FR2_PU2211A/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING') OR ('//EM_FR2_FLT_B/PUMP_SEL.CV' = 2 AND '//FR2_PU2211B/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING')) REM active during feed pump start	OPEN	NO
FR2_V2222A	Condition 1	Discharge to Hot Shock	'//FR2_DISCHARGE/OLEIN_DISC.CV' = 0	OPEN	NO
FR2_V2222A	Condition 2	Discharge to Tank Farm	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 1) AND ('//FR2_V2222B/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2222A	Condition 3	Discharge to 2HM211A	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 2) AND ('//FR2_V2223A/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2222A	Condition 4	Discharge to 2HM211B	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 3) AND ('//FR2_V2223B/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2222A	Condition 6	STOP is clicked	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 5) AND ('//FR2_PU2221/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2222B	Condition 1	Discharge to Tank Farm	'//FR2_DISCHARGE/OLEIN_DISC.CV' = 1	OPEN	NO
FR2_V2222B	Condition 2	Discharge to Hot Shock	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//FR2_V2222A/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2222B	Condition 3	Discharge to 2HM211A	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 2) AND ('//FR2_V2223A/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

PT. Wilmar Nabati Indonesia - DCS Serang Project

Revision 0 – 2019-Aug-16 – Page 88 of 92

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_V2222B	Condition 4	Discharge to 2HM211B	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 3) AND ('//FR2_V2223B/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2222B	Condition 6	STOP is clicked	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 5) AND ('//FR2_PU2221/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2223A	Condition 1	Discharge to 2HM211A	'//FR2_DISCHARGE/OLEIN_DISC.CV' = 2	OPEN	NO
FR2_V2223A	Condition 2	Discharge to Hot Shock	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//FR2_V2222A/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223A	Condition 3	Discharge to Tank Farm	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 1) AND ('//FR2_V2222B/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223A	Condition 4	Discharge to 2HM211B	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 3) AND ('//FR2_V2223B/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223A	Condition 6	STOP is clicked	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 5) AND ('//FR2_PU2221/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2223B	Condition 1	Discharge to 2HM211B	'//FR2_DISCHARGE/OLEIN_DISC.CV' = 3	OPEN	NO
FR2_V2223B	Condition 2	Discharge to Hot Shock	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//FR2_V2222A/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223B	Condition 3	Discharge to Tank Farm	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 1) AND ('//FR2_V2222B/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223B	Condition 4	Discharge to 2HM211A	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 2) AND ('//FR2_V2223A/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2223B	Condition 6	STOP is clicked	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 5) AND ('//FR2_PU2221/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2224A	Condition 1	HotShock, Precool Enabled	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//TP01.CV' != 0) AND ('//FR2_V2224B/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 89 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_V2224A	Condition 2	HotShock, Precool Disabled	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//TP01.CV' = 0) AND ('//FR2_V2224C/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2224A	Condition 3	STOP/HOTSHOCK not selected	('//FR2_DISCHARGE/OLEIN_DISC.CV' != 0)	CLOSE	NO
FR2_V2224B	Condition 1	HotShock, Precool Enabled	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//FR2_V2224A/TP01.CV' != 0)	OPEN	NO
FR2_V2224B	Condition 2	HotShock, Precool Disabled	('//FR2_DISCHARGE/OLEIN_DISC.CV' = 0) AND ('//FR2_V2224A/TP01.CV' = 0) AND ('//FR2_V2224A/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	CLOSE	NO
FR2_V2224B	Condition 3	STOP/HOTSHOCK not selected	('//FR2_DISCHARGE/OLEIN_DISC.CV' != 0) AND ('//FR2_V2224A/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	CLOSE	NO
FR2_V2230	Condition 1	LSL2230 ON	('//FR2_LSL2230/DI1/DISC_ACT.CV' = 1) AND ('//FR2_DISCHARGE/START.CV' = 1)	OPEN	NO
FR2_V2230	Condition 2	LSH2230 ON	('//FR2_LSH2230/DI1/DISC_ACT.CV' = 1) AND ('//FR2_DISCHARGE/START.CV' = 1)	CLOSE	NO
FR2_V2231(x)	Condition 1	Transfer to TankFarm	('//FR2_DISCHARGE/STEARIN_DISC_(x).CV' = 0) AND ('//FR2_V2233/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
FR2_V2231(x)	Condition 2	Circulation	('//FR2_DISCHARGE/STEARIN_DISC_(x).CV' = 1) AND ('//FR2_DISCHARGE/START_S_(x).CV' = 1)	OPEN	NO
FR2_V2231(x)	Condition 3	STOP is clicked	('//FR2_DISCHARGE/START_S_(x).CV' = 0) AND ('//FR2_PU2231(x)/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2232A	Condition 1	FR2_TT22312 >= TP01	('//FR2_TT2231C/PV.CV' >= '/TP01.CV') AND ('//FR2_DISCHARGE/START.CV' = 1)	CLOSE	NO
FR2_V2232A	Condition 2	FR2_TT22312 <= TP02	('//FR2_TT22312/PV.CV' <= '/TP02.CV') AND ('//FR2_DISCHARGE/START.CV' = 1)	OPEN	NO
FR2_V2232B	Condition 1	FR2_TT2231D >= TP01	('//FR2_TT2231D/PV.CV' >= '/TP01.CV') AND ('//FR2_DISCHARGE/START.CV' = 1)	CLOSE	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

PT. Wilmar Nabati Indonesia - DCS Serang Project

Revision 0 – 2019-Aug-16 – Page 90 of 92

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
FR2_V2232B	Condition 2	FR2_TT2231D <= TP02	('//FR2_TT2231D/PV.CV' <= 'TP02.CV') AND (('//FR2_DISCHARGE/START.CV' = 1))	OPEN	NO
FR2_V2233	Condition 1	TK231A/B TRANSFER	((('//FR2_DISCHARGE/STEARIN_DISC_A.CV' = 0) AND (('//FR2_DISCHARGE/START_S_A.CV' = 1))) OR ((('//FR2_DISCHARGE/STEARIN_DISC_B.CV' = 0) AND (('//FR2_DISCHARGE/START_S_B.CV' = 1)))	CLOSE	NO
FR2_V2233	Condition 2	TK231A switch to Stop/Circulation	((('//FR2_DISCHARGE/STEARIN_DISC_A.CV' = 1) AND (('//FR2_V2231A/PV_D.CV' = '_EDC_VLV_PV:OPEN')) AND ((('//FR2_DISCHARGE/START_S_B.CV' = 0) AND (('//FR2_PU2231B/PV_D.CV' = '_EDC_MTR2_PV:STOPPED'))	CLOSE	NO
FR2_V2233	Condition 3	TK231B switch to Stop/Circulation	((('//FR2_DISCHARGE/STEARIN_DISC_B.CV' = 1) AND (('//FR2_V2231B/PV_D.CV' = '_EDC_VLV_PV:OPEN')) AND ((('//FR2_DISCHARGE/START_S_A.CV' = 0) AND (('//FR2_PU2231A/PV_D.CV' = '_EDC_MTR2_PV:STOPPED'))	CLOSE	NO
FR2_V2233	Condition 4	TK231A/B switch to CIRCULATE	((('//FR2_DISCHARGE/STEARIN_DISC_A.CV' = 1) AND (('//FR2_V2231A/PV_D.CV' = '_EDC_VLV_PV:OPEN')) AND	CLOSE	NO



Unit Functional Design Specification - Fractionation #2

3258585-PSS-PAS-FSP-00004

Revision 0 – 2019-Aug-16 – Page 91 of 92

PT. Wilmar Nabati Indonesia - DCS Serang Project

3258585

Module	Name	Description	Expression	Force SP State	Delay On (s)
			((!//FR2_DISCHARGE/STEARIN_DISC_B.CV' = 1) AND (!//FR2_V2231B/PV_D.CV' = 'EDC_VLV_PV:OPEN'))		
FR2_V2233	Condition 5	TK231A & B STOP	(!//FR2_DISCHARGE/START_S_A.CV' = 0) AND (!//FR2_DISCHARGE/START_S_B.CV' = 0) AND (!//FR2_PU2231A/PV_D.CV' = 'EDC_MTR2_PV:STOPPED') AND (!//FR2_PU2231B/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
FR2_V2244	Condition 1	LSHH241 CLEARED & PU241 STOP	(!//FR2_LSHH2241/DISC_ALM.CUALM' = 0) AND (!//FR2_PU2241/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2252	Condition 1	Core Oil CIRCULATION	(!//FR2_DISCHARGE/COREOIL_DISC.CV' = 0)	OPEN	NO
FR2_V2252	Condition 2	Core Oil TRANSFER	(!//FR2_DISCHARGE/COREOIL_DISC.CV' = 1) AND (!//FR2_V2253/PV_D.CV' = 'EDC_VLV_SP:OPEN')	CLOSE	NO
FR2_V2252	Condition 3	STOP is clicked	(!//FR2_DISCHARGE/COREOIL_DISC.CV' = 2) AND (!//FR2_PU2251/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2253	Condition 1	Core Oil TRANSFER	!//FR2_DISCHARGE/COREOIL_DISC.CV' = 1	OPEN	NO
FR2_V2253	Condition 2	Core Oil CIRCULATION	(!//FR2_DISCHARGE/COREOIL_DISC.CV' = 0) AND (!//FR2_V2252/PV_D.CV' = 'EDC_VLV_SP:OPEN')	CLOSE	NO
FR2_V2253	Condition 3	STOP is clicked	(!//FR2_DISCHARGE/COREOIL_DISC.CV' = 2) AND (!//FR2_PU2251/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
FR2_V2264(x)	Condition 1	PT22612(x) > Core Blow Pressure	(!//FR2_PT22612(x)/PV.CV' > !//EM_FR2_FLT_(x)/TP019_VALUE.CV') AND ((!//EM_FR2_FLT_(x)/A_TARGET.CV' = 3) OR	OPEN	NO



Module	Name	Description	Expression	Force SP State	Delay On (s)
			('//EM_FR2_FLT_(x)/A_TARGET.CV' = 5) OR ('//EM_FR2_FLT_(x)/A_TARGET.CV' = 10)) AND ('//EM_FR2_FLT_(x)/A_STATE.CV' = 50)) REM active during CORE BLOW 1,2 & ADJUST SQZ FOR Core		
FR2_V2264(x)	Condition 2	PT22612(x) <= Core Blow Pressure	('//FR2_PT22612(x)/PV.CV' <= '//EM_FR2_FLT_(x)/TP019_VALUE.CV') AND (((('//EM_FR2_FLT_(x)/A_TARGET.CV' = 3) OR ('//EM_FR2_FLT_(x)/A_TARGET.CV' = 5) OR ('//EM_FR2_FLT_(x)/A_TARGET.CV' = 10)) AND ('//EM_FR2_FLT_(x)/A_STATE.CV' = 50)) REM active during CORE BLOW 1,2 & ADJUST SQZ FOR Core	CLOSE	NO

Table 79: Module Force Setpoints

7 MODULE PERMISSIVE

7.1 Functional Description

The section list down the permissive configurations.

*(n) denotes Crystallizer number from 01 to 22

Module	Name	Description	Expression	Interlock State	Delay Off (s)	Delay On (s)
FR2_AG22(n)	Condition 1	LT22(n) < Agitator Start Level	'//FR2_LT22(n)/PV.CV' < '//EM_FR2_CRY(n)/TP015_VALUE.CV'	STOP	NO	NO

Table 80: Module Permissive